

# Convolution absorbing boundaries for explicit-circuit quantum simulation of the wave equation

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Explicit quantum circuits for the wave equation, built by Hamiltonian simulation, are restricted to closed domains, in which outgoing waves reflect off the edge of the computational region and return. We lift that restriction with absorbing boundaries of the convolution type—a complex-frequency-shifted perfectly matched layer realised through exponential-kernel memory variables—and make the resulting non-unitary dynamics quantum-implementable by Schrödingerisation. We obtain three results. First, explicit gate-level circuits for the absorbing evolution, obtained by extending a Bell-basis term-evolution circuit to projector-valued operator strings and Trotterising to second order; these run end to end on seventeen to twenty-one qubits and agree with exact references to within a tenth of a percent to a percent. Second, a structural obstruction: the memory form of the absorbing generator carries an irreducibly indefinite Hermitian part, whose largest eigenvalue grows as the square root of the absorption strength divided by the grid spacing and survives any diagonal rescaling of the memory fields. The known recovery threshold for Schrödingerisation then makes the post-selection cost grow exponentially in the simulated time. Third, a remedy: a Lyapunov symmetrizer, precomputed classically, renders the transformed generator dissipative and replaces that time-dependent penalty with a time-independent conditioning factor of several hundred, measured on grids of up to four thousand unknowns and saturating under mesh refinement. The crossover is early: past it the symmetrized recovery is cheaper by four to twenty-six orders of magnitude in post-selection cost, and at the longest horizons tested it is the only recovery that works.

## I. INTRODUCTION

Quantum algorithms for partial differential equations (PDEs) divide into oracle-based<sup>1</sup> constructions, which assume block-encoded or sparse-access operators [1, 2], and the explicit-circuit line initiated for hyperbolic problems by Sato *et al.* [3], in which every gate of the time-stepping circuit is written down and the cost is dominated by a number of elementary operations polynomial in the register size. Explicit-circuit constructions for wave-type equations published to date operate on *closed* domains: Dirichlet, Neumann, or periodic boundaries, for which the semi-discrete generator is skew-Hermitian and the evolution is genuinely unitary [3, 4]; the extension to nonconservative systems [5] admits a bulk damping term  $\zeta(x) \partial_t u$  via a linear combination of Hamiltonian simulations, but its boundaries remain closed walls and its demonstrations are confined to the undamped acoustic and uniform heat equations. Scattering and radiation problems, however, are posed on open domains; truncating the domain with hard walls reflects every outgoing wave back into the region of interest. The same requirement propagates into wave-based inverse problems: full waveform inversion repeats the forward solve on a truncated domain at every iteration, and hybrid quantum-classical workflows for it—in which parameterized quantum circuits are embedded inside a domain-decomposed physics-informed network and trained end to end [6]—

inherit the boundary problem unchanged. That route is variational and targets near-term hardware, whereas the explicit-circuit programme pursued here targets fault-tolerant hardware with provable gate counts; both need a forward model that does not reflect. Classical solvers handle this with absorbing layers, the de facto standard being the perfectly matched layer (PML) and its convolutional, complex-frequency-shifted variant (CPML) [7–10]. Absorption is dissipation: the generator acquires a non-trivial Hermitian part, and the dynamics ceases to be unitary.

Schrödingerisation [11] is a general route for running such non-unitary linear dynamics on a quantum computer: the warped phase-space transform turns  $\dot{z} = Az$  into a genuine Hamiltonian evolution on one extra register, at the price of a post-selected recovery slice. Figure 1 summarises the resulting four-step structure of this work.

The programme now spans artificial boundary conditions for the time-dependent Schrödinger equation [12], inhomogeneous terms and the recovery-threshold theory [13], ill-posed problems [14], Maxwell’s equations [15, 16], elastic waves [17], gate-level circuits for parabolic problems [18, 19], query-optimal formulations [20], and implicit-explicit integrators for multiscale dynamics including the bulk-damped telegraph equation [21]. What it does not yet contain is an absorbing boundary for a classical wave-type equation, at either matrix or circuit level.

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<sup>1</sup> oracle is an unitary black box, you can run but do not need to know what is inside

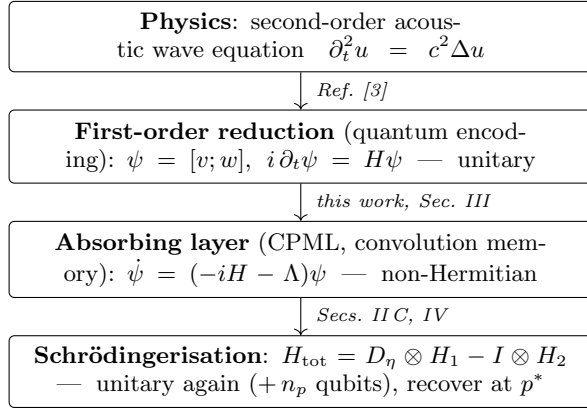


FIG. 1. The four-step structure: second-order acoustic physics, first-order quantum encoding, convolution absorbing layer applied in that first-order form, and Schrödingerisation restoring unitarity. The cost of step three—the indefiniteness quantified by Lemma 1—and its remedy (the Lyapunov symmetrizer) are the subject of Sec. III D.

### A. Related work and positioning

The positioning below was verified by a full-text literature review through June 2026.<sup>2</sup>

*a. The core niche is open.* No published or preprinted work through June 2026 combines any absorbing boundary (PML, CPML, complex absorbing potential, sponge, or Dirichlet-to-Neumann map) for *classical wave-type equations* with Schrödingerisation or any other unitarisation, at matrix or circuit level. Jin and Zhang [17] (elastic waves) state verbatim that non-periodic boundaries “remain an open issue” left “to the future.” Maxwell circuits exist only for perfect-electric-conductor boundaries [16] and for impedance boundaries at matrix level [15]; the 2026 circuit papers [22, 23] (Robin/Neumann/Dirichlet) contain no absorbing-boundary content. Two 2025 works corroborate this independently. Bösch *et al.* [4], simulating lossless wave equations by Hamiltonian simulation, state that owing to their “fundamentally nonenergy conserving nature” absorbing boundaries and PMLs are “not straightforward to implement...within the HS framework,” and resort to approximate *random* boundaries that scramble, rather than absorb, outgoing energy. Sato *et al.* [5] admit spatially varying bulk damping  $\zeta(x) \geq 0$  in an LCHS formulation, but the resulting Hermitian part is semidefinite by construction—the diagonal sponge/CAP case that Lemma 1 explicitly excludes—no layer-localised damping or absorbing-layer construction appears, and no damped-wave circuit is demonstrated.

*b. PML plus Schrödingerisation per se is prior art* for the time-dependent Schrödinger equation [12], at query-complexity level and without circuits. Our contribution is therefore framed as *wave-type equations plus explicit circuits*.

*c. The recovery threshold is established theory* [13, Thm. 3.1], and an indefinite boundary-augmented Hermitian part was already handled through that threshold for the heat equation [19, Remark 1]. Our delta is the *structural* result: Lemma 1 proves that the indefiniteness of CPML memory forms cannot be removed by any diagonal rescaling and quantifies its rate,  $\Theta(\sqrt{\sigma_{\max} c/\hbar})$ .

*d. Diagonal similarity preconditioning is prior art.* Liang and Liu [22] use a diagonal transform to enforce a positive-semidefinite Hermitian part before applying a linear-combination-of-Hamiltonian-simulation algorithm to parabolic problems. This is the foil that sharpens our symmetrizer contribution: Lemma 1 proves that diagonal transforms are *insufficient* for CPML generators; the full (non-diagonal) Lyapunov symmetrizer of Sec. III D is both necessary and sufficient.

*e. Bell-basis circuits for non-unitary PDEs are prior art* [18, 19]. Our circuit delta is narrower and is framed accordingly: projector-valued operator strings for separated multi-field (block) Hamiltonians, the two-level symmetrised splitting, and the absorbing-boundary application at 17–21 qubits (Sec. IV).

### B. Contributions

Ordered by verified novelty:

1. the first absorbing boundaries for wave-type equations in any unitarisation framework, closing the open problem stated in Ref. [17];
2. the threshold lemma (Lemma 1) and the Lyapunov symmetrizer that circumvents it, with the diagonal-insufficiency contrast to Ref. [22];
3. end-to-end validated explicit circuits, extending the Bell-basis constructions of Refs. [3, 18] to projector-valued block structure, with gold-standard classical validation.

## II. BACKGROUND

### A. The wave equation as Hamiltonian simulation

Following Ref. [3], the wave equation  $\partial_t^2 u = c^2 \Delta u$  is first-order-reduced through  $\psi = [v; w]$  with  $v = \partial_t u$  and  $w = ic \partial_x u$  (discretely,  $w = icDu$ ), so that  $i \partial_t \psi = H \psi$  with

$$H = \begin{bmatrix} 0 & D \\ D^\dagger & 0 \end{bmatrix}, \quad (1)$$

<sup>2</sup> Adversarially cross-checked sweep over published and preprinted work, 2023–2026; all primary claims below were confirmed against the full texts of the cited papers.

where  $D$  is the forward-difference operator and  $D^\dagger = -D_b$  its negative backward counterpart. Mixed Dirichlet/Neumann boundary conditions emerge from the operator products (the conventions of Ref. [3], verified numerically in our implementation). In two dimensions we use the separated-block form  $\psi = [v; w_x; w_y]$ —one block per field, in the style of Eq. (32) of Ref. [3] with  $d = 3$ —instead of the complex-combined two-block form, because per-direction damping requires separated gradient components.

### B. The convolutional perfectly matched layer

The CPML stretches each coordinate by the complex-frequency-shifted (CFS) factor

$$s(\omega) = \kappa + \frac{\sigma}{\alpha + i\omega}, \quad (2)$$

which in the time domain replaces every spatial derivative by a causal convolution with an exponential kernel [7],

$$\partial_x f \longrightarrow \frac{1}{\kappa} \partial_x f + \zeta * \partial_x f, \quad \zeta(t) = -\frac{\sigma}{\kappa^2} e^{-(\sigma/\kappa + \alpha)t} \theta(t). \quad (3)$$

Each convolution is carried by one memory field obeying the local ODE

$$\partial_t \phi = -\left(\frac{\sigma}{\kappa} + \alpha\right) \phi - \frac{\sigma}{\kappa^2} \partial_x f. \quad (4)$$

The damping profile is polynomially graded inside a layer of  $n_{\text{pml}}$  points,  $\sigma(d) = \sigma_{\text{max}}(d/L)^m$  with  $\sigma_{\text{max}} = -(m+1)c \ln R_0/(2L)$  [9]; we use the standard recipe  $m = 2$ ,  $R_0 = 10^{-3}$  with the parameter conventions of Komatitsch and Martin [8]. The profile is sampled at the staggered locations where each field effectively lives ( $v$  at integer nodes,  $w$  at half-cells). This stagger is load-bearing: an ablation shows that unstaggered sampling degrades the measured reflection by factors of 120–235.

### C. Schrödingerisation

Write the semi-discrete absorbing dynamics as  $\dot{z} = Az$  and split  $A = H_1 + iH_2$  with  $H_1 = (A + A^\dagger)/2$  and  $H_2 = (A - A^\dagger)/(2i)$ , both Hermitian<sup>4</sup>. The warped transform  $w(t, p) = e^{-p} z(t)$  (for  $p \geq 0$ , extended by a profile  $g$  below) obeys  $\partial_t w = -H_1 \partial_p w + iH_2 w$ , and a Fourier series in  $p$  decouples it into unitary evolutions per mode [11],

$$\hat{w}_k(T) = e^{-iT(\eta_k H_1 - H_2)} \hat{w}_k(0), \quad (5)$$

i.e. a Hamiltonian simulation of  $H_{\text{tot}} = D_\eta \otimes H_1 - I \otimes H_2$  on the system register plus one  $p$  register of  $n_p$  qubits. The initial data  $w(0, p) = g(p) z(0)$  is a product state across the two registers. The solution is recovered from a slice,

$$z(T) = e^{p^*} w(T, p^*), \quad p^* \geq \lambda_{\text{max}}^+(H_1) T, \quad (6)$$

where the threshold on  $p^*$  is Theorem 3.1 of Ref. [13]. The kinked profile  $g(p) = e^{-|p|}$  is only  $O(\Delta p)$  accurate; the  $C^1$  cubic profile of [13, Remark 4.9] restores  $O(\Delta p^2)$  while keeping  $g(p) = e^{-p}$  for  $p \geq 0$ , leaving Eq. (6) unchanged.

## III. ABSORBING GENERATORS

### A. One dimension: exact collapse of the convolution

With  $\kappa = 1$ ,  $\alpha = 0$  the convolution collapses *exactly* (for data supported outside the layer) to block-diagonal damping<sup>5</sup>,

$$\dot{\psi} = A\psi, \quad A = -iH - \text{diag}(\sigma_v, \sigma_w), \quad (7)$$

so  $H_1 = -\text{diag}(\sigma) \preceq 0$  and the recovery threshold (6) is unconditional ( $\lambda^+ = 0$ , any small  $p^* > 0$  is valid). The proof is the memory ansatz  $\phi = -\sigma v/\gamma$ , which satisfies the memory ODE (4) identically for data supported outside the layer; we verify the collapse numerically to  $6 \times 10^{-13}$ . The general CFS memory form (two extra block qubits) is provided for  $\kappa > 1$  or  $\alpha > 0$ .

### B. Two dimensions: memory form and classical validation

In 2D the damping acts per direction, so the state uses the separated-block form padded to eight blocks (three extra block qubits),

$$[v; w_x; w_y; \phi_{vx}; \phi_{vy}; \phi_x; \phi_y; 0], \quad (8)$$

with the memory injections masked to the layer strips. Corners need no special treatment—both directions' memory fields are simultaneously active, the structural advantage of unsplit CPML over split formulations [7]. Long-time stability is a serious concern for PMLs in general [10, 24, 25]; here we measure  $\max \text{Re eig}(A) = 0$  to round-off across all tested configurations.

*Gold-standard validation.* Comparing the truncated domain against a  $4\times$  enlarged reference, the true reflection is  $3.4 \times 10^{-4}$  (1D,  $n_{\text{pml}} = 12$ ) and  $2.8 \times 10^{-4}$  (2D  $32 \times 32$ ,  $n_{\text{pml}} = 8$ ), at the floor reported by Komatitsch and Martin [8].

<sup>3</sup> This is the a linear ODE, need to find  $z(t)$  given initial condition  $z(0)$ ;  $z$  is Psi in this case

<sup>4</sup> Correct for all squared  $A$

<sup>5</sup> Sato:  $A = -iH$ , closed system

*Convergence* [1D gold standard,  $N = 128$  versus an  $N = 512$  reference, plateau over  $T \in \{100, 120, 140\}$  with plateau spread  $< 10^{-9}$ , i.e. fully converged; Figs. 2(a,b)]:

- versus  $n_{\text{pml}}$  at  $R_0 = 10^{-3}$ :  $6.7 \times 10^{-4}$  (4),  $2.6 \times 10^{-4}$  (6),  $2.6 \times 10^{-4}$  (8),  $3.4 \times 10^{-4}$  (12),  $4.2 \times 10^{-4}$  (16)—*non-monotonic*, with a minimum at  $n_{\text{pml}} \approx 6$ –8. The two contributions are the discrete transition error, which decreases with layer width, and the wall-return reflection, which *increases toward the design value from below*: node sampling of the polynomial profile overestimates  $\int \sigma dx$  by a factor  $1 + O(1/n_{\text{pml}})$ , so the effective round-trip reflection is  $R_{\text{eff}} = R_0^{\Sigma \sigma h / f \sigma}$  ( $R_0^{1.38} = 7.1 \times 10^{-5}$  at  $n_{\text{pml}} = 4$  rising to  $R_0^{1.09} = 5.2 \times 10^{-4}$  at 16); for  $n_{\text{pml}} \geq 8$  the measured reflection equals this prediction within 5–20%, i.e. the layer performs at design, converging to  $R_0$  from below [Fig. 2(a)]. Jointly rescaling the staggered profile pair so the discrete sum matches the design integral (calibration=‘discrete’, Appendix D) makes the realised reflection flat at 0.81–1.08  $R_0$  over the whole sweep—a one-line calibration fix that applies to classical CPML codes as well;
- versus  $R_0$  at  $n_{\text{pml}} = 12$ :  $4.5 \times 10^{-3}$  ( $10^{-2}$ ),  $3.4 \times 10^{-4}$  ( $10^{-3}$ ),  $4.3 \times 10^{-5}$  ( $10^{-4}$ )—tracking the design reflection  $R_0$  within a factor  $\sim 2.3$  over two decades. The Collino–Tsogka amplitude formula [9] is predictive in this discretisation.

### C. The recovery-threshold obstruction

The memory form pays a price. The injection column ( $\gamma$  on one side) and the memory-source row ( $\sigma \partial_x / \gamma$  on the other) of each memory field are structurally asymmetric, and this asymmetry cannot be scaled away. Throughout this section  $\beta_j = \sigma_j / \kappa_j + \alpha_j \geq 0$  denotes the CFS decay rate at node  $j$ ; we set the wave speed  $c = 1$  (so the stencil entries obey  $|D_{ji}| = 1/h$ ), the general rate restoring a factor of  $c$ .

**Lemma 1 (diagonal irreducibility)** *Let  $A$  be the memory-form CPML generator of Sec. IIIB and let  $G = \text{diag}(\gamma_j) > 0$  be any diagonal rescaling of the memory fields. Let  $s_j$  denote the magnitude of the memory-source coupling ( $s_j \sim \sigma_j / (\gamma_j h)$  for the difference stencil). Then the Hermitian part  $H_1(G)$  of the rescaled generator satisfies*

$$\lambda_{\max}(H_1(G)) \geq \max_j \max\{f(\gamma_j, \beta_j), g(s_j, \beta_j)\} > 0, \quad (9)$$

with  $f(\gamma, \beta) = (\sqrt{\beta^2 + \gamma^2} - \beta)/2$ , arising from the  $2 \times 2$  principal submatrix  $\begin{bmatrix} 0 & \gamma/2 \\ \gamma/2 & -\beta \end{bmatrix}$  on a  $(v_j, \phi_j)$  pair by Cauchy interlacing, and  $g(s, \beta) = f(s, \beta)$  from a  $(w_i, \phi_j)$  pair (the one-directional coupling entry  $s$  enters  $H_1$  as

$s/2$ , so both pairs are governed by the same  $f$ ). Optimising  $\gamma$  against the two bounds gives

$$\lambda_{\max}(H_1) = \Theta\left(\sqrt{\sigma_{\max} c/h}\right) \quad \text{as } h \rightarrow 0, \quad (10)$$

independent of the scaling.

The full proof is given in Appendix A. Its consistency check is instructive: the collapsed 1D form of Sec. IIIB has *no* injection pair (the memory is eliminated and the damping is diagonal), so the lemma does not apply—and indeed  $H_1 = -\text{diag}(\sigma) \leq 0$  there.

*Consequence.* By the threshold of Eq. (6) [13, Thm. 3.1], recovery requires  $p^* \geq \lambda^+ T$  with  $\lambda^+ \equiv \lambda_{\max}^+(H_1)$ , and the post-selection probability of the recovery window scales as  $e^{-2p^*} = e^{-2\lambda^+ T}$ . Numerically: at  $\sigma_{\max} = 0.5$  on an  $8 \times 8$  grid,  $\lambda^+ = 0.683$ ; recovery below the threshold is  $O(1)$  wrong, while above it the error drops to  $2 \times 10^{-3}$  relative ( $10^{-4}$  with  $n_p = 11$ ); see Fig. 3. The sponge/CAP alternative (diagonal damping  $\sigma_x + \sigma_y$ , four blocks) has  $H_1 \preceq 0$  unconditionally at the price of matched-layer quality.

*CPML versus sponge at matched resources.* A matched-resource sweep quantifies this tradeoff: both absorbers on the same  $32 \times 32$  grid with equal layer widths  $n_{\text{pml}} \in \{4, 8, 12\}$  and default parameters, each measured against one shared  $4 \times$  enlarged ( $128 \times 128$ ) hard-wall reference by the gold-standard protocol of Sec. IIIB, the reflection being the maximum interior error over  $T \in \{8, 16, 24, 32, 40\}$  (Table I). At equal layer budget the CPML reaches its discrete wall-return level of  $2.7 \times 10^{-4}$  by  $n_{\text{pml}} = 8$  [consistent with the  $2.8 \times 10^{-4}$  gold-standard value of Sec. IIIB], while the sponge stays at the percent level, improving only from  $6.2 \times 10^{-2}$  to  $3.6 \times 10^{-2}$  across the sweep—a quality gap of a factor of 170 at  $n_{\text{pml}} = 8$  (130 at 12). The quantum-side cost inverts the ranking. The sponge has  $\lambda^+ = 0$  exactly, so its recovery post-selects at the  $T$ -independent  $e^{2p^*} \approx 1.2$  (default slice  $p^* \approx 0.11$ ); the CPML pays  $e^{2\lambda^+ T}$ , with  $\lambda^+ = 0.93$ –1.54 tracking the  $\Theta(\sqrt{\sigma_{\max} c/h})$  rate of Lemma 1 to within 5%—already  $7 \times 10^7$  at  $T = 8$  and  $1.6 \times 10^{39}$  at  $T = 40$  for  $n_{\text{pml}} = 8$ . Neither absorber dominates at matched resources: the sponge trades two orders of magnitude in reflection for an essentially free recovery, and the CPML trades an exponentially growing post-selection penalty for matched-layer accuracy—precisely the penalty that the symmetrizer of Sec. IIID replaces by a  $T$ -independent conditioning factor.

### D. The Lyapunov symmetrizer

A matrix  $W > 0$  solving the Lyapunov inequality  $WA + A^\dagger W \preceq 0$  gives, with  $S = W^{1/2}$ , a transformed generator  $\tilde{A} = SAS^{-1}$  whose Hermitian part satisfies

$$\tilde{H}_1 = \frac{1}{2} S^{-1} (WA + A^\dagger W) S^{-1} \preceq 0 \quad (11)$$

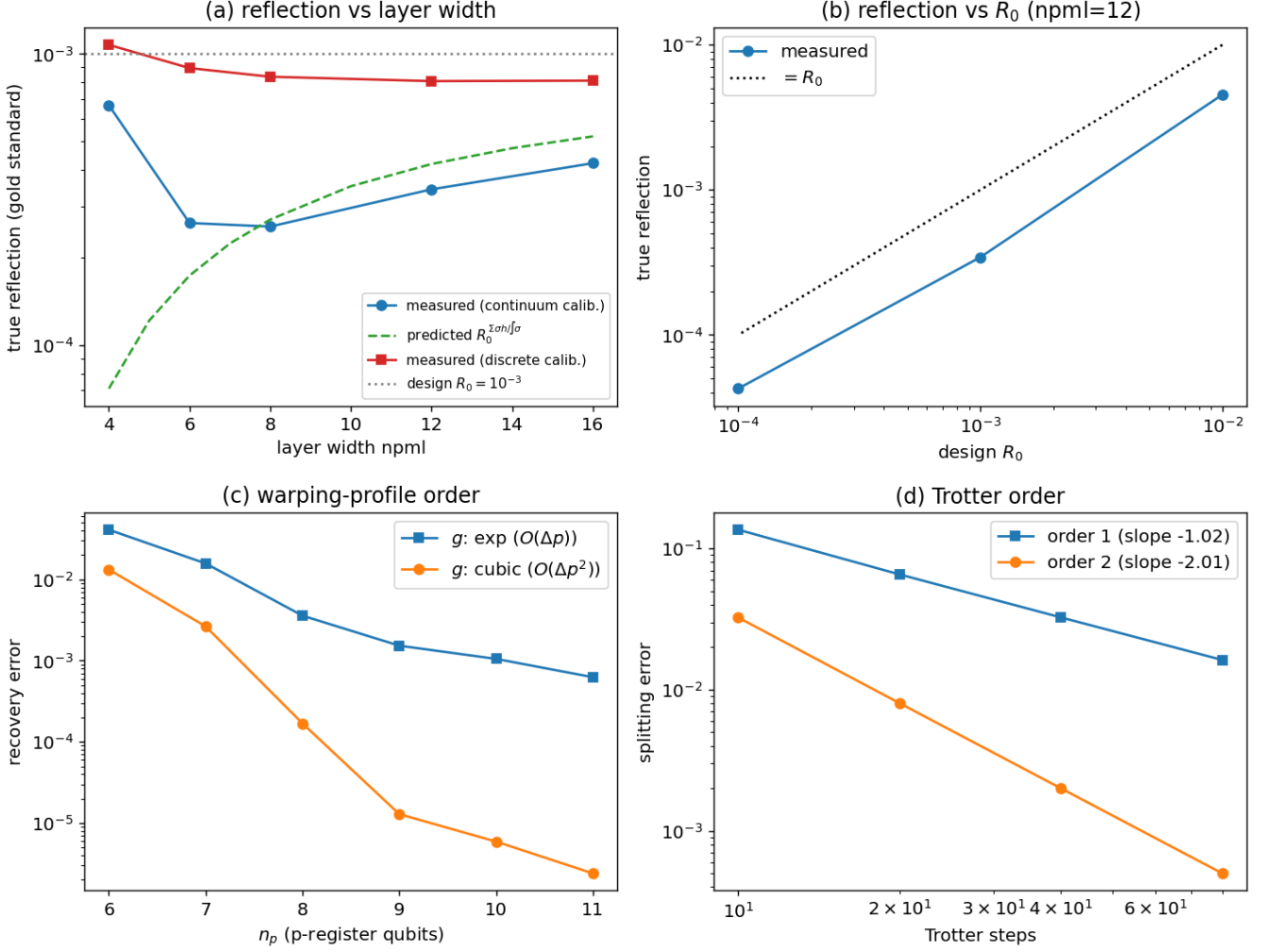


FIG. 2. Classical and Schrödingerisation convergence. (a) Gold-standard reflection (truncated domain versus  $4\times$  reference) versus layer width  $n_{\text{pml}}$  at design reflection  $R_0 = 10^{-3}$ . The curve is non-monotonic: the transition error falls with layer width while the wall return rises toward the design  $R_0$  from below, because node sampling of the polynomial profile over-damps thin layers (dashed curve: the predicted discrete wall-return  $R_{\text{eff}} = R_0^{\Sigma\sigma h/\int\sigma}$ , which the measurement tracks within 5–20% for  $n_{\text{pml}} \geq 8$ ). Red squares: the discretely calibrated profile pair (Appendix D), for which the realised reflection is flat at the design value. (b) Reflection versus  $R_0$  at  $n_{\text{pml}} = 12$ : the measured reflection tracks the Collino–Tsogka design value within a factor  $\sim 2.3$  over two decades. (c) Recovery error versus  $p$ -register size  $n_p$  for the kinked  $e^{-|p|}$  warping profile [ $O(\Delta p)$ ] and the  $C^1$  cubic profile [ $O(\Delta p^2)$ ] of Ref. [13], 1D CPML benchmark. (d) Trotter splitting error versus step count at orders 1 and 2 (fitted slopes  $-1.02$  and  $-2.01$ ), measured against the exact Schrödingerised evolution to isolate the splitting error.

by congruence. Recovery of  $\tilde{z} = Sz$  is then valid at any  $p^* > 0$ , and  $z = S^{-1}\tilde{z}$  is recovered classically (or folded into the readout). Since  $A$  is only marginally stable (interior modes are neutral), we construct  $W$  from the  $\epsilon$ -shifted generator:

$$A_\epsilon = A - \epsilon I, \quad A_\epsilon^\dagger W + W A_\epsilon = -I, \quad S = W^{1/2}, \quad (12)$$

the shift being undone classically by the factor  $e^{\epsilon T}$ . An exact identity follows (Proposition 1 in Appendix E: congruence gives  $\text{Herm}(S A_\epsilon S^{-1}) = -W^{-1}/2$  exactly; verified to better than  $10^{-11}$  relative in all 18 measured con-

figurations): the transform of the *original*  $A$  has

$$\lambda_{\max}(\tilde{H}_1) = \epsilon - \frac{1}{2\lambda_{\max}(W)} \approx \epsilon, \quad (13)$$

while the transform of  $A_\epsilon$  has  $\tilde{H}_1 \preceq 0$  strictly. We therefore use the shifted transform; the undo factor  $e^{\epsilon T} \leq e^{0.4}$  over the horizons tested is trivial.

*Measured results* [grids  $8 \times 8$ ,  $16 \times 8$ ,  $16 \times 16$  ( $n = 512$ – $2048$ ),  $\sigma_{\max} \in \{0.5, 1.0\}$ ,  $\epsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ ; Lyapunov residuals  $\leq 1.4 \times 10^{-14}$ ; Tables II and III, Fig. 4]:

- *End-to-end recovery at the default small  $p^*$*  ( $\epsilon = 10^{-3}$ ): transformed errors  $1.2 \times 10^{-6}$ – $3.5 \times 10^{-5}$  on

TABLE I. CPML versus sponge at matched resources:  $32 \times 32$  grid, equal layer width  $n_{\text{pml}}$ , default parameters [ $m = 2$ ,  $R_0 = 10^{-3}$ , Collino–Tsogka amplitudes  $\sigma_{\text{max}} = 2.59/1.30/0.86$  at  $n_{\text{pml}} = 4/8/12$ ]. Reflection is the gold-standard interior error against a single shared  $4 \times$  enlarged ( $128 \times 128$ ) hard-wall reference, maximised over  $T \in \{8, 16, 24, 32, 40\}$  ( $\|z_0\| = 1$ ).  $\lambda^+ = \lambda_{\text{max}}^+(H_1)$  sets the CPML post-selection penalty  $e^{2\lambda^+ T}$  (quoted at  $T = 40$ ) and tracks  $\sqrt{\sigma_{\text{max}}}$  to within 5%; the sponge has  $\lambda^+ = 0$ , post-selecting at the  $T$ - and  $n_{\text{pml}}$ -independent  $e^{2p^*} = 1.2$  for the default recovery slice  $p^* = 0.11$ .

| $n_{\text{pml}}$ | Reflection (max over $T$ ) |                      | CPML recovery cost |                             |
|------------------|----------------------------|----------------------|--------------------|-----------------------------|
|                  | CPML                       | sponge               | $\lambda^+(H_1)$   | $e^{2\lambda^+ T}$ , $T=40$ |
| 4                | $4.2 \times 10^{-3}$       | $6.2 \times 10^{-2}$ | 1.54               | $2.2 \times 10^{53}$        |
| 8                | $2.7 \times 10^{-4}$       | $4.7 \times 10^{-2}$ | 1.13               | $1.6 \times 10^{39}$        |
| 12               | $2.8 \times 10^{-4}$       | $3.6 \times 10^{-2}$ | 0.93               | $1.5 \times 10^{32}$        |

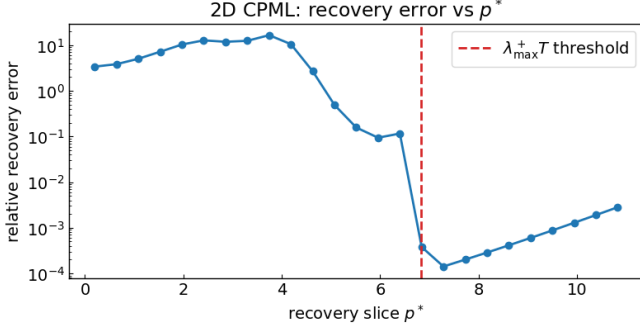


FIG. 3. The recovery threshold in action (2D CPML,  $8 \times 8$  grid,  $\sigma_{\text{max}} = 0.5$ ,  $T = 10$ ). Relative recovery error versus the slice position  $p^*$ : below the threshold  $p^* = \lambda^+ T$  (dashed line,  $\lambda^+ = 0.683$ ) the recovered state is  $O(1)$  wrong; just above it the error drops by four orders of magnitude, then grows slowly as  $e^{p^*}$  amplifies the  $p$ -grid discretisation floor.

the configurations with  $n_p \in \{10, 11\}$ , essentially flat in  $T$  over  $T \in \{10, 20, 40\}$ ; the  $16 \times 16$  run with a reduced  $p$  register ( $n_p = 8$ ) gives  $1.1 \times 10^{-4}$ – $3.8 \times 10^{-3}$ , consistent with its  $O(\Delta p^2)$  floor. The untransformed system at the same  $p^*$  returns  $O(1)$ – $O(10)$  garbage growing with  $T$ .

- *The direct above-threshold method is worse than its nominal  $e^{2\lambda^+ T}$  cost:* at  $T = 20$  it degrades (errors 0.24–0.45 at  $\sigma_{\text{max}} = 0.5$ ) and at  $T = 40$  it explodes (error up to  $7 \times 10^{15}$  at  $p^* = 38$ ) because  $e^{p^*}$  amplifies the  $p$ -grid floor. At long  $T$  the symmetrizer is not merely cheaper—it is the only working recovery.
- *Conditioning:*  $\kappa_2(S) = 4.0 \times 10^2$ – $8.6 \times 10^2$  at  $\epsilon = 10^{-2}$  (the recommended setting), flat in  $\sigma_{\text{max}}$ , scaling as  $\sim \epsilon^{-0.6 \text{ to } -1.0}$ —and saturating in  $n$ : extending the measurement to  $n = 4096$  gives  $\kappa_2(S) = 564.5$  at  $\epsilon = 10^{-2}$  versus 566.9 at  $n = 2048$  (local slope  $-0.006$ ; over the full range  $n = 512$ – $4096$  the

fitted power-law exponent drops from 0.25 to 0.16 with degrading fit quality, i.e. a power law is the wrong model because  $\kappa$  is flattening).  $\lambda^+$  itself saturates ( $0.683 \rightarrow 0.694$  over  $8 \times$  in  $n$ ) and  $\lambda_{\text{min}}(W)$  stays at 0.66–0.68, so the  $\kappa$  growth is driven entirely by  $\lambda_{\text{max}}(W) \sim 1/(2\epsilon)$ : the conditioning is  $\epsilon$ -controlled, not grid-controlled, and the symmetrizer cost does not blow up under mesh refinement. The eigenvector symmetrizer  $W = (VV^\dagger)^{-1}$  is four to six orders of magnitude worse conditioned ( $\kappa_2 \sim 3 \times 10^8$ ) and fails outright on near-defective configurations—the Lyapunov route is the right construction.

- *Tradeoff:*  $T^* = \ln \kappa^2 / (2\lambda^+) = 6.7$ – $16.0$  across all configurations; at  $T = 20$  the symmetrizer wins by 4–9 orders of magnitude in post-selection cost, at  $T = 40$  by 16–26 orders.  $T^*$  grows only logarithmically with  $\kappa$ , hence very weakly with grid size.

*Structure and compressibility of  $S$ .* We measured the structure of  $S$  directly and tested 25 approximate symmetrizers (distance-banded, magnitude-thresholded, block-pattern, identity-plus-low-rank, hybrid, and inverse-side-banded families) for dissipativity and end-to-end recovery on the  $n = 512$  and  $n = 1024$  systems. Three findings. (i) Identity-plus-low-rank is *ruled out*: the correction  $E = S - \alpha I$  is numerically full rank (995 of 1024 singular values above  $10^{-2}$  of the top), and  $\lambda_{\text{max}}(\tilde{H}_1)$  stays at the *untransformed*  $\lambda^+ = 0.67$  for ranks 8–64—no few-term LCU implementation of  $S$  exists. (ii) Strict sparsification preserving  $\lambda_{\text{max}}(\tilde{H}_1) \leq \epsilon$  requires essentially the full active support ( $\approx 48$ – $51\%$  of entries; on the remaining “dead” indices—interior memory and padding, 30–50% of the space— $S$  is exactly a multiple of the identity, which is free structure). (iii) *Moderate compression works* once the residual  $\lambda_b$  of an approximate  $S_b$  is absorbed as an additional shift (undone by  $e^{\lambda_b T} \leq e^{2.6} \approx 14$  at the worst measured  $\lambda_b = 0.066$ ,  $T = 40$ ): magnitude thresholding at 35% density recovers to  $8.7 \times 10^{-6}$  ( $T = 10$ ) and  $5.4 \times 10^{-5}$  ( $T = 40$ ); a rank-32-plus-band-8 hybrid at 26% reaches  $1.3 \times 10^{-5}$  and  $6.6 \times 10^{-4}$  (all  $n = 1024$ ; the conditioning of every usable  $S_b$  stays  $\sim 5 \times 10^2$ , equal to the exact  $S$ ). The energy of  $E$  is 99.6% concentrated on layer-strip index pairs. The implementation route is therefore block-encoding of a  $\sim 35\%$ -dense operator (times the free dead-index identity factor), not a local circuit or a short LCU; at current theory the symmetrizer is best used as classical pre/post-processing, and a structured-ansatz Lyapunov construction (solving for  $W$  within a polynomially parametrised operator family) is the open problem these measurements sharpen.

*Honest limitations.* The construction requires a dense  $O(n^3)$  classical precomputation of  $W$ ,  $S$ ,  $S^{-1}$ ; the fate of the sparse quantised-tensor-train (QTT) term structure under the dense similarity (Trotter circuits for  $\hat{A}$ ) is not assessed; the  $\kappa$  scaling beyond  $n = 4096$  is extrapolation (though saturating); the compression study is at  $n \leq$

TABLE II. Lyapunov-symmetrizer conditioning across all measured configurations [grids  $(n_x, n_y)$  qubits per dimension, i.e.  $8 \times 8$ ,  $16 \times 8$ ,  $16 \times 16$  points;  $n$  is the full state dimension including the eight blocks].  $\lambda^+ = \lambda_{\max}^+(H_1)$  is the indefiniteness of the untransformed generator;  $\kappa_2(S) = \sqrt{\text{cond}_2(W)}$  the conditioning of the symmetrizer;  $\lambda_{\max}(\tilde{H}_1^{(\epsilon)})$  the (strictly negative) largest Hermitian-part eigenvalue of the  $\epsilon$ -shifted transform  $SA_\epsilon S^{-1}$ ; and  $T^* = \ln \kappa^2/(2\lambda^+)$  the crossover time beyond which the symmetrizer beats the direct above-threshold recovery. All Lyapunov residuals are  $\leq 1.4 \times 10^{-14}$  (relative).

| Grid           | $n$  | $\sigma_{\max}$ | $\epsilon$ | $\lambda^+(H_1)$ | $\kappa_2(S)$     | $\lambda_{\max}(\tilde{H}_1^{(\epsilon)})$ | $T^*$ |
|----------------|------|-----------------|------------|------------------|-------------------|--------------------------------------------|-------|
| $8 \times 8$   | 512  | 0.5             | $10^{-2}$  | 0.683            | $4.0 \times 10^2$ | $-4.6 \times 10^{-6}$                      | 8.8   |
| $8 \times 8$   | 512  | 0.5             | $10^{-3}$  | 0.683            | $3.0 \times 10^3$ | $-8.1 \times 10^{-8}$                      | 11.7  |
| $8 \times 8$   | 512  | 0.5             | $10^{-4}$  | 0.683            | $1.1 \times 10^4$ | $-6.0 \times 10^{-9}$                      | 13.6  |
| $8 \times 8$   | 512  | 1.0             | $10^{-2}$  | 0.927            | $5.1 \times 10^2$ | $-5.2 \times 10^{-6}$                      | 6.7   |
| $8 \times 8$   | 512  | 1.0             | $10^{-3}$  | 0.927            | $2.9 \times 10^3$ | $-1.6 \times 10^{-7}$                      | 8.6   |
| $8 \times 8$   | 512  | 1.0             | $10^{-4}$  | 0.927            | $9.8 \times 10^3$ | $-1.4 \times 10^{-8}$                      | 9.9   |
| $16 \times 8$  | 1024 | 0.5             | $10^{-2}$  | 0.687            | $5.1 \times 10^2$ | $-3.0 \times 10^{-6}$                      | 9.1   |
| $16 \times 8$  | 1024 | 0.5             | $10^{-3}$  | 0.687            | $7.0 \times 10^3$ | $-1.5 \times 10^{-8}$                      | 12.9  |
| $16 \times 8$  | 1024 | 0.5             | $10^{-4}$  | 0.687            | $3.1 \times 10^4$ | $-8.0 \times 10^{-10}$                     | 15.0  |
| $16 \times 8$  | 1024 | 1.0             | $10^{-2}$  | 0.956            | $7.5 \times 10^2$ | $-2.4 \times 10^{-6}$                      | 6.9   |
| $16 \times 8$  | 1024 | 1.0             | $10^{-3}$  | 0.956            | $7.0 \times 10^3$ | $-2.8 \times 10^{-8}$                      | 9.3   |
| $16 \times 8$  | 1024 | 1.0             | $10^{-4}$  | 0.956            | $2.7 \times 10^4$ | $-1.8 \times 10^{-9}$                      | 10.7  |
| $16 \times 16$ | 2048 | 0.5             | $10^{-2}$  | 0.694            | $5.7 \times 10^2$ | $-2.4 \times 10^{-6}$                      | 9.1   |
| $16 \times 16$ | 2048 | 0.5             | $10^{-3}$  | 0.694            | $9.9 \times 10^3$ | $-7.7 \times 10^{-9}$                      | 13.3  |
| $16 \times 16$ | 2048 | 0.5             | $10^{-4}$  | 0.694            | $6.6 \times 10^4$ | $-1.7 \times 10^{-10}$                     | 16.0  |
| $16 \times 16$ | 2048 | 1.0             | $10^{-2}$  | 0.972            | $8.6 \times 10^2$ | $-1.9 \times 10^{-6}$                      | 6.9   |
| $16 \times 16$ | 2048 | 1.0             | $10^{-3}$  | 0.972            | $1.2 \times 10^4$ | $-8.7 \times 10^{-9}$                      | 9.7   |
| $16 \times 16$ | 2048 | 1.0             | $10^{-4}$  | 0.972            | $6.3 \times 10^4$ | $-3.4 \times 10^{-10}$                     | 11.4  |

TABLE III. End-to-end recovery tradeoff: direct versus symmetrized recovery, errors  $\|z_{\text{rec}} - e^{AT} z_0\|_2 / \|e^{AT} z_0\|_2$ . “Small  $p^*$ ” is the library-default slice three grid points above zero. For the untransformed system above threshold,  $p^* = \lambda^+ T + 1$  and the post-selection penalty is  $e^{2p^*}$ ; entries marked “–” exceed the  $p$ -domain ( $\lambda^+ T + 1 > p_{\max}$ ). The symmetrized column uses the  $\epsilon$ -shifted transform at  $\epsilon = 10^{-3}$  recovered at the same small  $p^*$ , with the  $T$ -independent penalty  $\kappa_2^2(S)$  of Table II. The  $16 \times 16$  run uses a reduced  $p$  register ( $n_p = 8$ ), which sets its higher error floor.

| Configuration                                        | $T$ | Untransformed       |                        |                      |                      | Symmetrized ( $\epsilon = 10^{-3}$ ) |
|------------------------------------------------------|-----|---------------------|------------------------|----------------------|----------------------|--------------------------------------|
|                                                      |     | err. (small $p^*$ ) | $p^* \geq \lambda^+ T$ | err. (above thr.)    | penalty $e^{2p^*}$   | err. (small $p^*$ )                  |
| $8 \times 8$ ( $\sigma_{\max} = 0.5$ , $n_p = 11$ )  | 10  | 3.4                 | 7.8                    | $1.2 \times 10^{-4}$ | $6.3 \times 10^6$    | $1.2 \times 10^{-6}$                 |
|                                                      | 20  | 8.3                 | 14.7                   | $2.4 \times 10^{-1}$ | $5.4 \times 10^{12}$ | $2.4 \times 10^{-6}$                 |
|                                                      | 40  | $2.6 \times 10^1$   | –                      | –                    | –                    | $4.4 \times 10^{-6}$                 |
| $8 \times 8$ ( $\sigma_{\max} = 1.0$ , $n_p = 11$ )  | 10  | 7.2                 | 10.3                   | $2.2 \times 10^{-2}$ | $8.4 \times 10^8$    | $2.1 \times 10^{-5}$                 |
|                                                      | 20  | $1.6 \times 10^1$   | 19.6                   | $3.0 \times 10^2$    | $1.0 \times 10^{17}$ | $2.3 \times 10^{-5}$                 |
|                                                      | 40  | $2.9 \times 10^1$   | 38.1                   | $7.3 \times 10^{15}$ | $1.2 \times 10^{33}$ | $3.5 \times 10^{-5}$                 |
| $16 \times 8$ ( $\sigma_{\max} = 0.5$ , $n_p = 10$ ) | 10  | 2.5                 | 7.9                    | $1.1 \times 10^{-4}$ | $7.4 \times 10^6$    | $4.2 \times 10^{-6}$                 |
|                                                      | 20  | 4.8                 | 14.8                   | $4.5 \times 10^{-1}$ | $6.7 \times 10^{12}$ | $6.0 \times 10^{-6}$                 |
|                                                      | 40  | $1.3 \times 10^1$   | –                      | –                    | –                    | $1.1 \times 10^{-5}$                 |
| $16 \times 16$ ( $\sigma_{\max} = 0.5$ , $n_p = 8$ ) | 10  | 2.0                 | 8.0                    | $1.3 \times 10^{-2}$ | $9.2 \times 10^6$    | $1.1 \times 10^{-4}$                 |
|                                                      | 20  | 5.6                 | 15.0                   | $7.5 \times 10^1$    | $1.0 \times 10^{13}$ | $5.2 \times 10^{-4}$                 |
|                                                      | 40  | $2.5 \times 10^1$   | –                      | –                    | –                    | $3.8 \times 10^{-3}$                 |

1024.

## IV. EXPLICIT CIRCUITS

### A. Projector-extended Bell-basis term evolution

The circuit of Sato *et al.* [3] exponentiates ladder-string terms (alphabet  $\{I, \sigma_{01}, \sigma_{10}\}$ ) via a Bell-basis

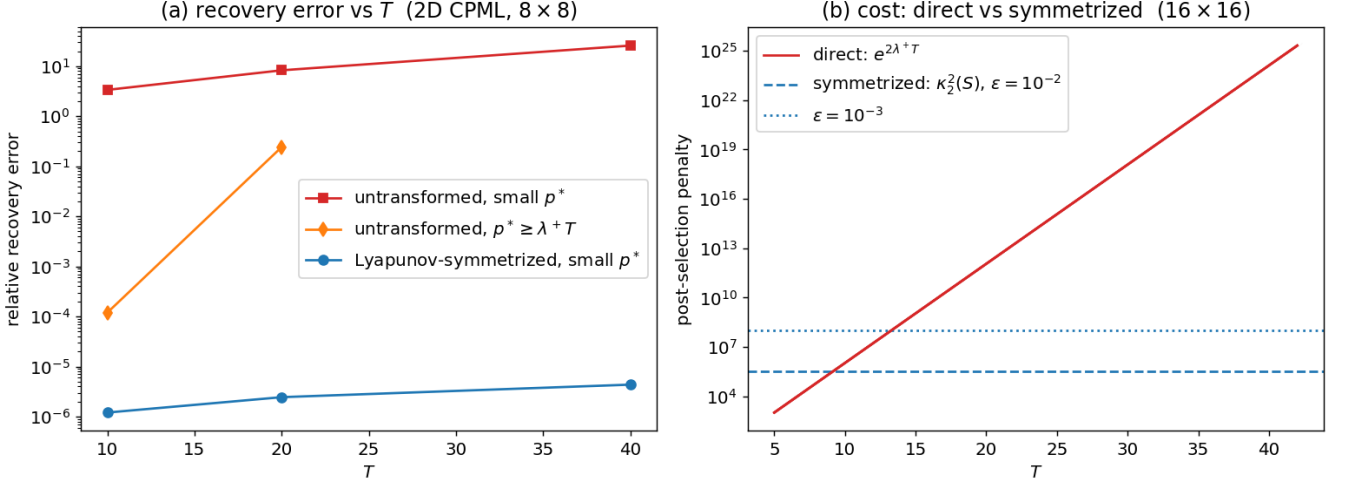


FIG. 4. The Lyapunov symmetrizer. (a) Relative recovery error versus evolution time  $T$  for the 2D CPML system ( $8 \times 8$  grid,  $\sigma_{\max} = 0.5$ ): untransformed recovery at the default small  $p^*$  (squares) is  $O(1)$  garbage; untransformed recovery above the threshold  $p^* \geq \lambda^+ T$  (diamonds) degrades with  $T$  as  $e^{p^*}$  amplifies the  $p$ -grid floor and leaves the admissible  $p$ -domain entirely by  $T = 40$ ; the Lyapunov-symmetrized recovery at the same small  $p^*$  (circles,  $\epsilon = 10^{-3}$ , shifted transform) stays at the  $10^{-6}$  level, essentially flat in  $T$ . (b) Post-selection cost on the  $16 \times 16$  grid: the direct  $e^{2\lambda^+ T}$  penalty (solid) against the  $T$ -independent symmetrizer penalty  $\kappa_2^2(S)$  at  $\epsilon = 10^{-2}$  (dashed) and  $10^{-3}$  (dotted); the crossovers are the  $T^*$  values of Table II.

change followed by one multi-controlled  $R_z$ ; Hu, Jin, Liu, and Zhang [18] extended the framework to non-unitary PDE dynamics via Schrödingerisation, and Jin, Liu, and Yu [19] built gate-level circuits for the heat equation with physical boundaries. Relative to these we add the *projector* extension: strings containing  $z = |0\rangle\langle 0|$  and  $o = |1\rangle\langle 1|$  (block projectors of separated multi-field systems) enter as plain 0-/1-controls; the Bell rotation outside the projected subspace is undone exactly, so the circuit implements  $e^{-i\delta t(cP+\text{h.c.})}$  for projector-valued  $P$ . Figure 5(a) shows the compiled circuit for a representative projector-containing term of the separated 2D wave Hamiltonian, and Fig. 5(b) the full Schrödingerised pipeline it slots into.

Due diligence (full-text check of both papers): neither Ref. [18] nor Ref. [19] circuitises projector-containing strings—both use only the  $\{I, \sigma_{01}, \sigma_{10}\}$  alphabet. We confirmed this at code level in the published companion notebook of Ref. [18]: its term circuit ( $W_j$ ) implements the contiguous ladder pattern with rotations controlled uniformly on all lower qubits (an all-or-nothing  $X$ -conjugation flag serving the periodic corner terms), with no per-qubit 0-/1-control selection (the nilpotent shift needs no projector guards in scalar single-field problems, and Ref. [19] routes boundary data into source terms precisely to avoid non-Toeplitz structure). Ref. [5] comes closest: its symbolic operator algebra over  $\{I, \sigma_{00}, \sigma_{11}, \sigma_{01}, \sigma_{10}\}$  produces projector factors from boundary-truncated shifts and piecewise-constant coefficients, and poly-depth Trotterisation is asserted by citation to Ref. [3]; but no gate-level compilation rule for the projector factors and no projector-term circuit

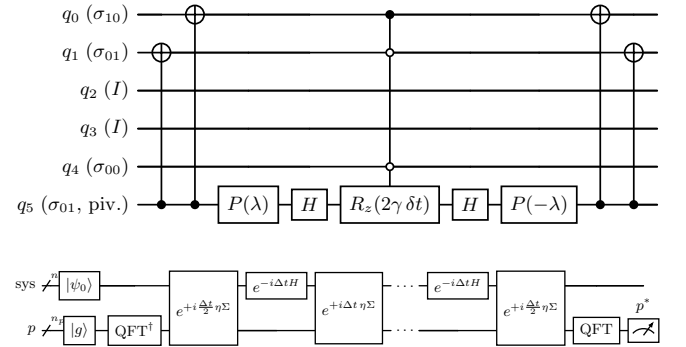


FIG. 5. Explicit circuits. (a) Bell-basis evolution  $e^{-i\delta t(cP+\text{h.c.})}$  of the projector-containing term  $P = \sigma_{01} \otimes \sigma_{00} \otimes I \otimes I \otimes \sigma_{01} \otimes \sigma_{10}$  of the separated 2D wave Hamiltonian (string `mzIImp`, qubit  $q_k$  carrying string character  $5 - k$ ;  $\lambda = \arg c$ ,  $\gamma = |c|$ ;  $\lambda = 0$  for the real wave-equation coefficients): CNOTs rotate the ladder qubits to the Bell basis, and the single multi-controlled  $R_z$  carries the block projector  $\sigma_{00}$  ( $q_4$ ) as an *open-circle* 0-control—the entire compilation rule of Sec. IV A. The gate sequence is unitarily identical to the library-compiled circuit (verified to machine precision). (b) The Schrödingerised pipeline (Sec. IV B) on bundled registers: product-state preparation  $|g\rangle \otimes |\psi_0\rangle$ ,  $QFT^\dagger$  on the  $p$  register, Strang-split steps alternating the Bell wave circuit  $e^{-i\Delta t H}$  with the diagonal damping phases  $e^{+i\Delta t\eta\Sigma}$  (half steps at the ends), closing  $QFT$ , and recovery by measuring the  $p$  register, post-selecting the slice at  $p^*$ , and rescaling classically by  $e^{p^*}$ .

is given. A caveat we state explicitly: the abstract class  $|a\rangle\langle b| + |b\rangle\langle a|$  over basis states in Lemma 1 of Refs. [3, 18]

implicitly *contains* Hermitian-paired projector strings as an existence statement; our contribution is the explicit alphabet extension and compilation rule (projector factor  $\rightarrow$  0-/1-control) and its application to multi-field block-coupled systems, not a new diagonalisation principle. The extension is backward compatible. Unitary-level validation: the error is pure per-step Trotter error  $0.50 \delta t^2$  with no phase offset. This unlocks the separated 2D wave Hamiltonian (zm/mz block couplings) needed for per-direction absorbing layers.

### B. The Schrödingerised circuit

The full pipeline is

$$\begin{aligned} \text{init}(g \otimes \psi_0) &\rightarrow \text{QFT}_p^\dagger \rightarrow [\text{Trotter steps}] \\ &\rightarrow \text{QFT}_p \rightarrow \text{slice at } p^*. \end{aligned} \quad (14)$$

Each step (diagonal  $H_1$ ) factorises into the wave circuit (the Bell construction of Sec. IV A on the system register) and a diagonal damping phase with entries  $e^{+i \delta t \eta_k \sigma_j}$ . Both Trotter levels are symmetrised (Strang outer; palindromic term order inner), with measured clean second order: errors  $4 \times 10^{-2} \rightarrow 10^{-2} \rightarrow 2.7 \times 10^{-3}$  at 50/100/200 steps on 17 qubits.

*Fitted orders* [1D benchmark, error versus the exact Schrödingerised evolution to isolate the splitting; Fig. 2(d)]: slope  $-1.02$  (order 1),  $-2.01$  (order 2 with both levels symmetrised); against the full matrix-exponential reference the order-2 error saturates at the  $p$ -discretisation floor, as expected. *Warping-profile order* [recovery error versus  $n_p$ , fixed 1D PML case; Fig. 2(c)]:  $e^{-|p|}$  slope  $\sim -1$  in  $\Delta p$  ( $4.1 \times 10^{-2} \rightarrow 6.3 \times 10^{-4}$  over  $n_p = 6\text{--}11$ );  $C^1$  cubic  $\sim -2$  ( $1.3 \times 10^{-2} \rightarrow 2.4 \times 10^{-6}$ ), confirming Remark 4.9 of Ref. [13] in this setting.

### C. End-to-end demonstrations

- 1D CPML, 14 qubits (6 system + 8  $p$ ), 60 steps,  $T = 30$ : recovered solution within  $7 \times 10^{-4}$  of the exact non-unitary evolution.
- 2D sponge,  $8 \times 8$  grid, 17 qubits,  $T = 10$ :  $2.7 \times 10^{-3}$  at 200 steps (Fig. 6).
- 2D sponge,  $32 \times 32$  grid, 21 qubits (12 system + 9  $p$ ), up to 240 steps (983 native operations per step): errors  $7.7 \times 10^{-3}$  /  $1.7 \times 10^{-2}$  /  $1.4 \times 10^{-2}$  at  $T = 8/16/24$ ; energy decays from 0.97 to  $1.25 \times 10^{-3}$  (absorption complete); nine minutes of statevector simulation (Fig. 7).

Qubit accounting: the absorbing overhead is +1 (sponge) or +2 (CPML) block qubits plus  $n_p \approx 8\text{--}10$  for the  $p$  register;  $n_p$  depends only logarithmically on  $T$ , the damping rate, and the target accuracy.

## V. COMPLEXITY

Term counts inherit the QTT structure of the difference operators:  $2n_x + 2$  terms in 1D and  $4n_x + 4$  in 2D ( $n_x = n_y$ ), exactly linear in the register size (fitted slopes  $4.000/2.000$ , intercepts  $4/2$ ). Transpiled to the  $\{\text{CX}, u\}$  basis, one Strang wave step costs  $217/459/834/1343$  operations for  $n_{\text{sys}} = 6/8/10/12$ ; a power-law fit gives exponent  $2.63$  with  $r^2 = 0.99997$  (Fig. 8)—convincingly polynomial, and in fact asymptotically quadratic: the term count is linear in  $n$  and the ancilla-free V-chain multi-controlled- $R_z$  is linear in its control count, so ops/step  $= O(n^2)$ . The constants are fixed by the measurements. The per-step cost in the  $\{\text{CX}, U\}$  basis is reproduced to  $0.03\%$  ( $r^2 = 0.9999999$ ) by the quadratic

$$G_{\text{step}}(n) = 16.7 n^2 - 112.7 n + 293, \quad (15)$$

whose structure is transparent: a Strang step exponentiates  $2n$  Hermitian-paired terms ( $4n_x + 4$  at  $n = 2n_x + 2$ ), each one Bell basis change plus one multi-controlled  $R_z$  whose ancilla-free V-chain cost is *exactly*  $32k - 31$  elementary operations for  $k \geq 8$  controls (measured), linear in  $k$ ; with the QTT ladder structure averaging  $\bar{k} \approx n/4$  controls per term, the leading coefficient is  $2n \cdot 32 \bar{k} / n^2 \approx 16.7$ . The total gate count is steps  $\times$  ops/step with steps  $= T/\delta t$  and  $\delta t$  set by the second-order bound  $\delta t = \sqrt{\varepsilon_{\text{tr}}/(C_2 T)}$ , where  $C_2$  is the Strang error constant of the slope- $(-2)$  fits of Fig. 2(d):

$$G_{\text{total}} = \sqrt{\frac{C_2 T^3}{\varepsilon_{\text{tr}}}} G_{\text{step}}(n) = O\left(\frac{T^{3/2}}{\sqrt{\varepsilon_{\text{tr}}}} n^2\right), \quad (16)$$

plus the damping diagonal per step and two  $\text{QFT}(n_p)$ ; the power-law fit across the measured range reads  $n^{2.63}$  because the subleading terms of Eq. (15) have not yet decayed at  $n \leq 12$ .

*Caveat (honest).* The damping factor is a single native **diagonal** instruction on the simulator, but a *generic* diagonal on  $n$  qubits synthesises to  $\Theta(2^n)$  elementary gates: we measure a CX count of exactly  $2^n - 2$  and totals within 15% of the  $2^{n+1} - 3$  synthesis bound. For hardware the structure must be exploited: the diagonal is  $D_\eta \otimes \Sigma$  with  $\eta$  linear in the  $p$  index and  $\sigma$  a degree- $m$  polynomial profile on the layer strips, so phase-polynomial constructions give  $\text{poly}(n)$  circuits, and the logic-minimisation synthesis of piecewise-constant diagonal operators of Ref. [5] (Espresso-compressed projector strings with matrix-product-state coefficient oracles) provides an existing  $\text{poly}(n)$  toolbox directly applicable to the strip-wise polynomial  $\sigma$  profiles; making this explicit is deferred to future work.

## VI. DISCUSSION AND OUTLOOK

The  $\lambda^+$  obstruction of Lemma 1 is generic for auxiliary-field absorbers—memory variables, auxiliary-differential-

### Gate-level absorbing 2D wave (sponge), 17 qubits

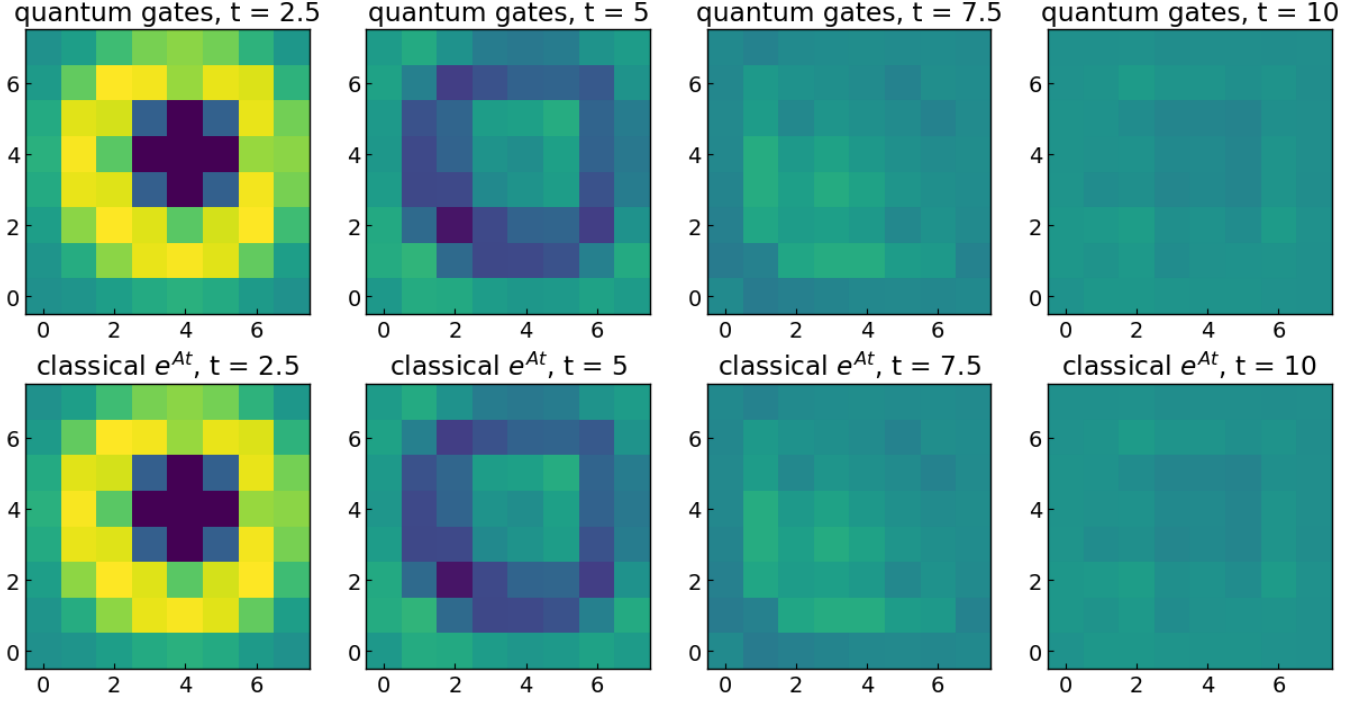


FIG. 6. Gate-level absorbing 2D wave evolution (sponge layer,  $8 \times 8$  grid, 17 qubits = 6 grid + 2 block + 9  $p$ -register). Top row: snapshots of the recovered field from the Trotterised quantum circuit at  $t = 2.5, 5, 7.5, 10$ . Bottom row: the exact non-unitary reference  $e^{At}\psi_0$ . The expanding pulse is absorbed in the boundary layer with no visible reflection; snapshot errors are at the  $10^{-3}$  level (Sec. IV C).

equation formulations, split fields—because the injection/source asymmetry is structural rather than an artefact of one discretisation. The Lyapunov symmetrizer is the corresponding general remedy, and its quantum-implementable variants (block-encoding  $S$ , structured or banded approximate symmetrizers) are open. Natural next steps include a hardware experiment in the style of Ref. [3]; quantum-signal-processing or qubitisation in place of Trotter (and the query-optimal Schrödingerisation formulation of Ref. [20]); CPML for Maxwell’s equations [15, 16]; and smoother warping profiles (error-function or numerically optimal) beyond the  $C^1$  cubic.

*Code and data availability.* The implementation (`lib/absorbing.py`, `lib/schrodingerisation.py`), all experiment scripts and JSON result files (`paper/experiments/`), and the executed demonstration notebook are available in the accompanying repository at <https://github.com/x-repos/qawave-cpml>.

### Appendix A: Proof of Lemma 1

Fix a layer node  $j$  with  $\sigma_j > 0$  and a stencil neighbour  $i$  such that the difference entry satisfies  $|D_{ji}| = 1/h$

(wave speed  $c = 1$ ; the general case restores a factor  $c$  in Eq. (10)). Write  $\beta = \sigma_j/\kappa_j + \alpha_j \geq 0$  for the CFS decay rate,  $\gamma$  for the injection coefficient, and  $s = \sigma_j|D_{ji}|/(\kappa_j^2\gamma)$  for the memory-source magnitude. Let  $G = \text{diag}(g) > 0$  be *any* positive diagonal similarity, and set

$$u = \frac{g_{v_j}}{g_{\phi_j}}, \quad t = \frac{g_{\phi_j}}{g_{w_i}}, \quad \rho = ut = \frac{g_{v_j}}{g_{w_i}}. \quad (\text{A1})$$

The transformed generator  $\tilde{A} = GAG^{-1}$  has unchanged diagonal, and the three relevant coupling families scale as follows.

- (i) *Injection pair*  $(v_j, \phi_j)$ : entry  $u\gamma$ , reverse entry 0, so  $\tilde{H}_1$  contains the principal  $2 \times 2$  submatrix

$$\begin{bmatrix} 0 & u\gamma/2 \\ \bar{u}\bar{\gamma}/2 & -\beta \end{bmatrix} \implies \lambda_{\max}(\tilde{H}_1) \geq f(|u|\gamma, \beta), \quad (\text{A2})$$

with  $f(x, \beta) := (\sqrt{\beta^2 + x^2} - \beta)/2$ .

- (ii) *Memory-source pair*  $(w_i, \phi_j)$ : entry  $ts$  analogously, giving  $\lambda_{\max}(\tilde{H}_1) \geq f(|t|s, \beta)$ .
- (iii) *Physical wave pair*  $(v_j, w_i)$ : elementwise,  $\tilde{A}[v, w] = -iD_{ji}\rho$  and  $\tilde{A}[w, v] = +iD_{ji}/\rho$  (using that  $D$  is

Gate-level absorbing 2D wave, 32x32 grid (21 qubits, 240 Trotter steps at T=24)

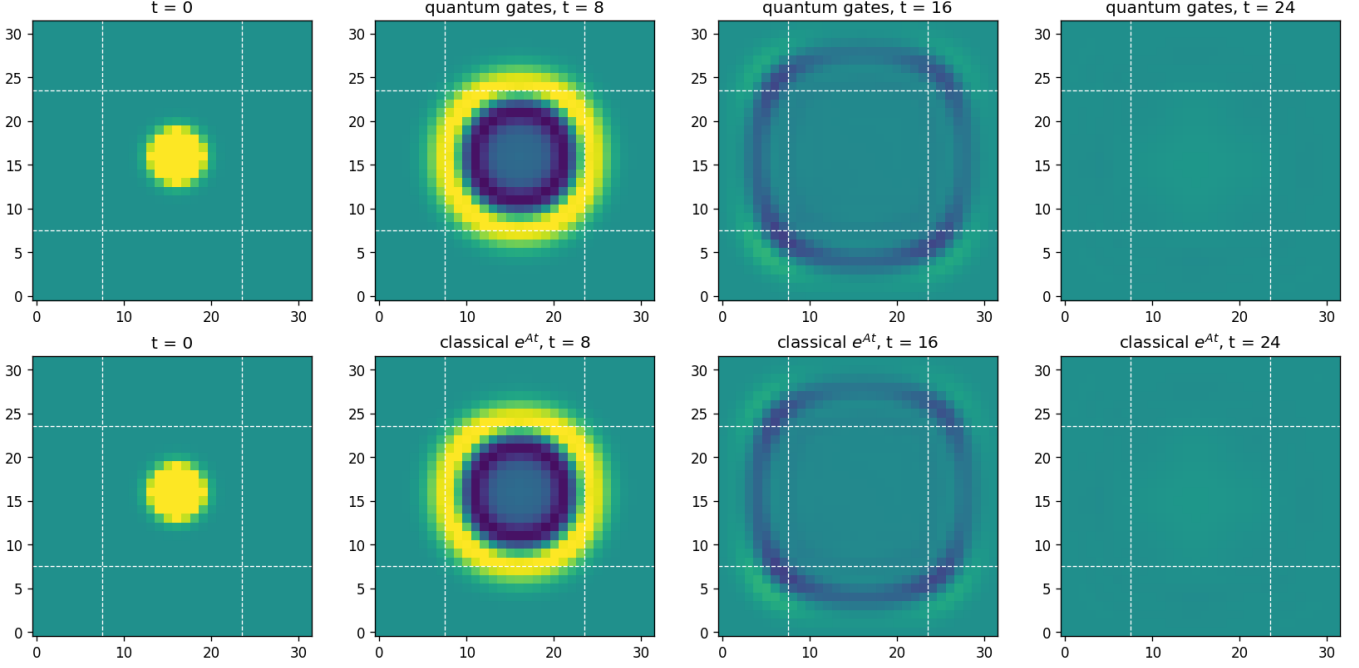


FIG. 7. Gate-level absorbing 2D wave evolution at scale:  $32 \times 32$  grid, 21 qubits (12 system + 9  $p$ -register), up to 240 second-order Trotter steps (983 native operations per step). Top row: recovered field from the quantum circuit at  $t = 0, 8, 16, 24$  (dashed lines mark the absorbing-layer strips). Bottom row: exact reference  $e^{At}\psi_0$ . The outgoing ring is absorbed across the layer; the total energy decays from  $0.97$  to  $1.25 \times 10^{-3}$ , and the recovered field matches the reference to  $7.7 \times 10^{-3}$ – $1.7 \times 10^{-2}$ .

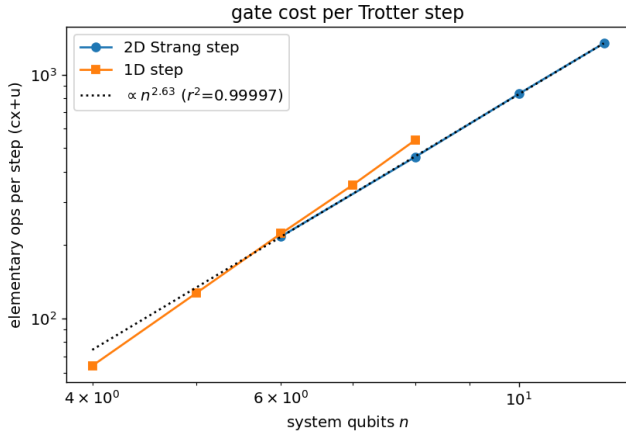


FIG. 8. Gate cost per Trotter step after transpilation to the  $\{CX, u\}$  basis, versus the number of system qubits  $n$ : 1D steps (squares) and 2D Strang steps (circles). The dotted line is the power-law fit  $\propto n^{2.63}$  ( $r^2 = 0.99997$ ) to the 2D series; the term count itself is exactly linear in  $n$  (Sec. V).

real and  $D^\dagger = -D_b$ ), so

$$\begin{aligned} \tilde{H}_1[v_j, w_i] &= \frac{iD_{ji}}{2}(\rho^{-1} - \rho) \\ \Rightarrow \lambda_{\max}(\tilde{H}_1) &\geq \frac{|D_{ji}|}{2}|\rho^{-1} - \rho|. \end{aligned} \quad (\text{A3})$$

All three are lower bounds on  $\lambda_{\max}(\tilde{H}_1)$  by Cauchy interlacing (the eigenvalues of any principal submatrix interlace those of the full Hermitian matrix).

*Case analysis.* If  $|\rho| \leq 1/2$  then  $|\rho^{-1} - \rho| \geq 3/2$ , so (iii) gives  $\lambda_{\max}(\tilde{H}_1) \geq 3/(4h)$ . Otherwise  $|\rho| > 1/2$  implies

$$|u\gamma| \cdot |t|s| = |\rho|\gamma s > \frac{\gamma s}{2} = \frac{\sigma_j |D_{ji}|}{2\kappa_j^2}, \quad (\text{A4})$$

hence  $\max(|u|\gamma, |t|s|) \geq \sqrt{\sigma_j/(2h\kappa_j^2)}$  and (i)–(ii) give  $\lambda_{\max}(\tilde{H}_1) \geq f(\sqrt{\sigma_j/(2h\kappa_j^2)}, \beta)$ . Therefore, for every positive diagonal scaling,

$$\lambda_{\max}(\tilde{H}_1) \geq \min \left\{ \frac{3}{4h}, \frac{\sqrt{\beta^2 + \sigma_j/(2h\kappa_j^2)} - \beta}{2} \right\}, \quad (\text{A5})$$

which is  $\Theta(\sqrt{\sigma_j/h})$  as  $h \rightarrow 0$  (the second branch is the minimum once  $\sigma_j/(2h) \gg \beta^2$ , i.e. for any fixed profile

and small  $h$ ). The bound is independent of  $\gamma$  and of the scaling  $G$ . The explicit balanced choice  $\gamma^* = \sqrt{2\sigma_{\max}/h}$  attains the same rate from above: measured  $\lambda^+ = 0.68$ – $0.97$  across our configurations, versus the lower bound  $0.18$  at the  $8 \times 8$ ,  $\sigma_{\max} = 0.5$  parameters—the bound is a rate, not a sharp constant. ■

*Consistency check.* The collapsed 1D form (Sec. III A) has no injection pair: the memory is eliminated and the damping is diagonal, so the lemma does not apply—and indeed  $H_1 = -\text{diag}(\sigma) \leq 0$  there.

## Appendix B: The convolutional PML in the Hamiltonian-simulation variables

This appendix expands the one-line statements of Secs. II B, III A, and III B: the causal kernel and memory ODE from the CFS stretch (Appendix B 1), the one-dimensional memory generator and the frequency-domain collapse (Appendix B 2), the proof that the collapse is exact (Appendix B 3), the two-dimensional eight-block generator (Appendix B 4), the staggered profile-sampling rule (Appendix B 5), and the profile formulas (Appendix B 6). Throughout,  $D$  is the forward-difference matrix and  $D^\dagger = -D_b$  its negative backward counterpart (Sec. II A); all operators are exactly those constructed in the accompanying code.

### 1. From the CFS stretch to the memory ODE

We fix the Fourier convention  $\hat{f}(\omega) = \int dt e^{-i\omega t} f(t)$ ,  $f(t) = (2\pi)^{-1} \int d\omega e^{+i\omega t} \hat{f}(\omega)$ , so that  $\partial_t \mapsto i\omega$  and the causal exponential has the transform  $\int_0^\infty dt e^{-\beta t} e^{-i\omega t} = (\beta + i\omega)^{-1}$  for  $\beta > 0$ . A PML stretches the coordinate into the complex plane,  $\tilde{x} = \int_0^x s(x', \omega) dx'$ , which in the frequency domain replaces every spatial derivative by  $\partial_x \mapsto s(\omega)^{-1} \partial_x$  inside the layer. For the CFS factor of Eq. (2), partial fractions in the variable  $i\omega$  give

$$\frac{1}{s(\omega)} = \frac{\alpha + i\omega}{\kappa(\alpha + i\omega) + \sigma} = \frac{1}{\kappa} + \hat{\zeta}(\omega), \quad (\text{B1})$$

with

$$\hat{\zeta}(\omega) = -\frac{\sigma}{\kappa^2} \frac{1}{\beta + i\omega}, \quad \beta := \frac{\sigma}{\kappa} + \alpha \quad (\text{B2})$$

the CFS decay rate: indeed  $1/s - 1/\kappa = (\kappa - s)/(\kappa s)$  with  $\kappa - s = -\sigma/(\alpha + i\omega)$  and  $\kappa s = \kappa^2(\beta + i\omega)/(\alpha + i\omega)$ . The single pole at  $i\omega = -\beta$  inverts, by the causal transform above, to the exponential kernel of Eq. (3),  $\zeta(t) = -(\sigma/\kappa^2) e^{-\beta t} \theta(t)$ , and frequency multiplication is time convolution:  $s^{-1} \partial_x f = \kappa^{-1} \partial_x f + \zeta * \partial_x f$ .

The convolution becomes local in time once it is named. Let  $\phi := \zeta * \partial_x f$ . Multiplying Eq. (B1) by  $(\beta + i\omega)$  gives the polynomial identity  $(\beta + i\omega) \hat{\zeta} = -\sigma/\kappa^2$ , hence  $i\omega \hat{\phi} = -\beta \hat{\phi} - (\sigma/\kappa^2) \widehat{\partial_x f}$ , which transforms back to the

memory ODE of Eq. (4). Equivalently, in the time domain,  $\partial_t \phi = \zeta(0^+) \partial_x f + \zeta' * \partial_x f$  with  $\zeta(0^+) = -\sigma/\kappa^2$  and  $\zeta' = -\beta \zeta$ . One first-order ODE per stretched derivative thus carries the convolution exactly; this is the recursive-convolution CPML of Roden and Gedney [7].

### 2. One dimension: memory generator and collapse

In the variables of Sec. II A the undamped 1D system is  $\dot{v} = -iDw$ ,  $\dot{w} = -iD^\dagger v$  [Eq. (1)]. Each equation contains exactly one spatial derivative, so the stretch generates one memory field per physical field. Sampling the profiles per field (Appendix B 5) defines the diagonal matrices  $\Sigma_v = \text{diag}(\sigma_{v,j})$ ,  $K_v = \text{diag}(\kappa_{v,j})$ ,  $B_v = \text{diag}(\sigma_{v,j}/\kappa_{v,j} + \alpha_{v,j})$ , and the layer projector  $\Pi_v = \text{diag}(\mathbf{1}[\sigma_{v,j} > 0])$ , and likewise for  $w$ . Applying the stretched-derivative rule to  $-iDw$  and carrying the convolution by the rescaled memory field  $\phi_v := \gamma^{-1} \zeta_v * (-iDw)$ —where  $\gamma > 0$  is a free constant rescaling under which the physical trajectory is invariant, set to the balanced value  $\gamma = \sqrt{2\sigma_{\max}/h}$  of Appendix A—gives, by Eq. (4),

$$\dot{v} = -iK_v^{-1} Dw + \gamma \phi_v, \quad \dot{\phi}_v = -B_v \phi_v + \frac{i}{\gamma} \Sigma_v K_v^{-2} Dw, \quad (\text{B3})$$

and the analogous pair for  $(w, \phi_w)$  with  $D \rightarrow D^\dagger$ ,  $v \leftrightarrow w$ . The source of  $\phi_v$  carries the factor  $\Sigma_v$ , so  $\phi_v$  vanishes identically outside the layer strips for memory-free data; masking the injection column by  $\Pi_v$  therefore changes no such trajectory while removing the spurious  $O(\gamma)$  interior entries that an unmasked column would contribute to the Hermitian part  $H_1$ . On the four-block state  $[v; w; \phi_v; \phi_w]$  (two extra block qubits) the generator is

$$A_{\text{mem}} = \begin{bmatrix} 0 & -iK_v^{-1}D & \gamma\Pi_v & 0 \\ -iK_w^{-1}D^\dagger & 0 & 0 & \gamma\Pi_w \\ 0 & \frac{i}{\gamma}\Sigma_v K_v^{-2}D & -B_v & 0 \\ \frac{i}{\gamma}\Sigma_w K_w^{-2}D^\dagger & 0 & 0 & -B_w \end{bmatrix}. \quad (\text{B4})$$

*Collapse for  $\kappa = 1$ ,  $\alpha = 0$ .* In this case the stretch degenerates:  $s(\omega) = (i\omega + \sigma)/(i\omega)$ , so the stretched  $v$  equation,  $i\omega \hat{v} = s_v(\omega)^{-1}(-iD\hat{w})$  per Fourier mode, multiplies out to

$$(i\omega + \sigma_v) \hat{v} = -iD\hat{w}, \quad (\text{B5})$$

because the symbol  $s(\omega)i\omega = i\omega + \sigma$  is a *first-degree polynomial* in  $i\omega$ . Transforming back,  $\dot{v} = -iDw - \sigma_v v$ , and likewise  $\dot{w} = -iD^\dagger v - \sigma_w w$ : the convolution collapses to the local damping of Eq. (7),  $A = -iH - \text{diag}(\sigma_v, \sigma_w)$ , with no memory fields at all.

### 3. Proof of the exact collapse

The frequency-domain argument above is formal (it commutes the stretch with the time-domain initial-value

problem); the following statement makes the equivalence exact at the semi-discrete level.

**Lemma 2 (exact collapse)** *Let  $\kappa \equiv 1$  and  $\alpha \equiv 0$  in the generator of Eq. (B4), so that  $B_v = \Sigma_v$  and  $B_w = \Sigma_w$ , and define the defects*

$$e_v := \phi_v + \gamma^{-1}\Sigma_v v, \quad e_w := \phi_w + \gamma^{-1}\Sigma_w w. \quad (\text{B6})$$

*Then  $\dot{e}_v = \dot{e}_w = 0$  along every solution of  $\dot{\Psi} = A_{\text{mem}}\Psi$ . Consequently a memory-form trajectory projects onto a trajectory of the collapsed generator Eq. (7) if and only if  $e_v(0) = e_w(0) = 0$ ; for the standard embedding  $\phi_v(0) = \phi_w(0) = 0$  this is the condition  $\Sigma_v v(0) = 0$  and  $\Sigma_w w(0) = 0$ , i.e. initial data supported outside the layer.*

*Proof.* Using both rows of Eq. (B3) with  $B_v = \Sigma_v$ ,

$$\begin{aligned} \dot{e}_v &= \left[ -\Sigma_v \phi_v + \frac{i}{\gamma} \Sigma_v D w \right] + \gamma^{-1} \Sigma_v \left[ -i D w + \gamma \Pi_v \phi_v \right] \\ &= (\Sigma_v \Pi_v - \Sigma_v) \phi_v = 0, \end{aligned}$$

since  $\Sigma_v \Pi_v = \Sigma_v$  (the projector is the support indicator of  $\Sigma_v$ ); identically for  $e_w$ . Equivalently, the block-row operator  $E = \begin{bmatrix} \gamma^{-1}\Sigma_v & 0 & I & 0 \\ 0 & \gamma^{-1}\Sigma_w & 0 & I \end{bmatrix}$  annihilates the generator from the left,  $EA_{\text{mem}} = 0$  (verified to round-off in our implementation). On the invariant slice  $e \equiv 0$  the memory ansatz  $\phi_v = -\gamma^{-1}\Sigma_v v$  of Sec. III A holds at all times, the injection term becomes  $\gamma \Pi_v \phi_v = -\Sigma_v v$ , and  $(v, w)$  obeys Eq. (7); conversely, every collapsed trajectory lifts to the memory form through the ansatz. ■

When  $e(0) \neq 0$ , the conserved defect acts on the physical equations as the static in-layer forcing  $\gamma \Pi_v e_v(0)$ ,  $\gamma \Pi_w e_w(0)$ , so the discrepancy between the two forms scales with the layer content  $\|\Sigma_v v(0)\|$ ,  $\|\Sigma_w w(0)\|$  of the initial data—and vanishes exactly for data supported outside the layer. Numerically, propagating the same interior pulse with Eq. (B4) and with Eq. (7) agrees to  $6 \times 10^{-13}$  (Sec. III A).

*Remark (the roles of  $\alpha$  and  $\kappa$ ).* The cancellation uses only  $\alpha \equiv 0$ : for a graded  $\kappa$  the modified defect  $e_v = \phi_v + \gamma^{-1}\Sigma_v K_v^{-1}v$  is conserved by the same computation, and the convolution collapses to the local form  $\dot{v} = -iK_v^{-1}Dw - K_v^{-1}\Sigma_v v$  (both statements again verified to round-off). For  $\alpha > 0$ , instead,  $\dot{e}_v = -\text{diag}(\alpha_v)\phi_v \neq 0$ —equivalently,  $s(\omega)i\omega$  is no longer polynomial in  $i\omega$ —and the memory fields are irreducible: the four-block form is then the genuine content of the CFS layer.

#### 4. Two dimensions: the eight-block generator

In 2D the damping acts per direction: the  $x$  stretch replaces every  $\partial_x$  and the  $y$  stretch every  $\partial_y$ . On the separated-block state of Sec. II A the undamped equations are  $\dot{v} = -iD_x w_x - iD_y w_y$ ,  $\dot{w}_x = -iD_x^\dagger v$ ,  $\dot{w}_y = -iD_y^\dagger v$ , with  $D_x = D \otimes I_{N_y}$  and  $D_y = I_{N_x} \otimes D$  on the flattened  $N_x N_y$  grid register ( $x$  register more significant). The  $v$  equation contains one derivative of each direction and acquires two memory fields  $\phi_{vx}$ ,  $\phi_{vy}$ ; each  $w$  equation contains a single derivative and acquires one,  $\phi_x$  for  $w_x$  and  $\phi_y$  for  $w_y$ . The four memory fields plus the three physical fields are padded by one zero block to the eight blocks of Eq. (8) (three block qubits).

Because  $\sigma_x$  depends on  $x$  only, its diagonal samples factorise on the grid register:  $\Sigma_v^x = S_x \otimes I_{N_y}$  and  $\Sigma_w^x = S_x^{\text{st}} \otimes I_{N_y}$ , where  $S_x = \text{diag } \sigma(x_j)$  holds the nodal samples and  $S_x^{\text{st}} = \text{diag } \sigma(x_{j-1/2})$  the half-cell samples (Appendix B5); symmetrically  $\Sigma_v^y = I_{N_x} \otimes S_y$  and  $\Sigma_w^y = I_{N_x} \otimes S_y^{\text{st}}$ . With  $\kappa \equiv 1$  (our 2D configurations), the decay rates are  $B_v^x = \Sigma_v^x + \text{diag } \alpha_x(x_j)$  etc., and the masks  $\Pi_v^x = \text{diag}(\mathbf{1}[\Sigma_v^x > 0])$ , ... are the layer-strip projectors. Repeating the derivation of Appendix B2 once per (field, stretched direction) pair gives the generator, in the block ordering of Eq. (8),

$$A = \begin{bmatrix} 0 & -iD_x & -iD_y & \gamma \Pi_v^x & \gamma \Pi_v^y & 0 & 0 & 0 \\ -iD_x^\dagger & 0 & 0 & 0 & 0 & \gamma \Pi_w^x & 0 & 0 \\ -iD_y^\dagger & 0 & 0 & 0 & 0 & 0 & \gamma \Pi_w^y & 0 \\ 0 & \frac{i}{\gamma} \Sigma_v^x D_x & 0 & -B_v^x & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{\gamma} \Sigma_v^y D_y & 0 & -B_v^y & 0 & 0 & 0 \\ \frac{i}{\gamma} \Sigma_w^x D_x^\dagger & 0 & 0 & 0 & 0 & -B_w^x & 0 & 0 \\ \frac{i}{\gamma} \Sigma_w^y D_y^\dagger & 0 & 0 & 0 & 0 & 0 & -B_w^y & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{on } [v; w_x; w_y; \phi_{vx}; \phi_{vy}; \phi_x; \phi_y; 0], \quad (\text{B7})$$

which is, entry for entry, the matrix assembled by the accompanying implementation.

*Strip masking.* The source of each memory field carries its own profile factor ( $\Sigma_v^x$  for  $\phi_{vx}$ , ...), so the field re-

TABLE IV. Staggered sampling of the damping profiles in 2D (forward-difference convention; the backward convention flips the half-cell shifts). Each profile is sampled on the grid of the field its equation updates.

| Field | Lives at           | Profile samples            |
|-------|--------------------|----------------------------|
| $v$   | $(x_j, y_k)$       | $\sigma(x_j), \sigma(y_k)$ |
| $w_x$ | $(x_{j-1/2}, y_k)$ | $\sigma(x_{j-1/2})$        |
| $w_y$ | $(x_j, y_{k-1/2})$ | $\sigma(y_{k-1/2})$        |

mains identically zero outside its layer strip for memory-free initial data; the corresponding injection column is masked by the strip projector, exactly as in 1D, so that  $H_1$  carries no spurious interior coupling.

*Corners.* In a corner cell both  $\sigma_x > 0$  and  $\sigma_y > 0$  hold simultaneously: both directions' memory fields are active there, the same generator entries of Eq. (B7) apply unchanged, and no corner-specific terms arise. This is the structural advantage of the unsplit convolutional PML over split-field formulations [7]. The measured stability,  $\max \text{Re eig}(A) = 0$  to round-off, is reported in Sec. III B.

### 5. The staggered sampling rule

For the forward-difference convention,  $(Dw)_j = (w_{j+1} - w_j)/h$ . The component  $v_j$  is updated through  $(Dw)_j$  and represents the field at the integer node  $x_j = jh$ ; the component  $w_j$  is updated through  $(D^\dagger v)_j \propto v_j - v_{j-1}$ , a difference centered at the half-cell point  $x_{j-1/2}$ , so the  $w$  field lives on the staggered grid (the backward convention shifts it to  $x_{j+1/2}$ ). The sampling rule follows: *every profile factor in an evolution equation is sampled on the grid of the field that the equation updates,*

$$\sigma_{v,j} = \sigma(x_j), \quad \sigma_{w,j} = \sigma(x_{j\mp 1/2}), \quad (\text{B8})$$

with the upper (lower) sign for the forward (backward) convention, and per direction in 2D (Table IV): only the  $x$  profile of the  $w_x$  family and the  $y$  profile of the  $w_y$  family are staggered; all other dependencies are nodal.

*Ablation.* Sampling all profiles at the integer nodes instead—the only change being the half-cell shift of the  $w$  profiles—degrades the measured gold-standard reflection of Sec. III B by factors of 120–235. At the reflection floors of Fig. 2(a,b) the half-cell stagger is load-bearing, not a refinement.

### 6. Profile formulas

Within a layer of  $n_{\text{pml}}$  points (width  $L = n_{\text{pml}}h$ ) the damping is polynomially graded in the depth  $d$  measured from the interface,

$$\sigma(d) = \sigma_{\text{max}} (d/L)^m, \quad 0 \leq d \leq L, \quad (\text{B9})$$

and zero outside; the interface node itself is undamped,  $\sigma(0) = 0$ , the outermost node attains  $\sigma_{\text{max}}$ , and staggered samples beyond the wall are clipped to  $d = L$ . The amplitude follows from the normal-incidence round trip: a wave crossing the layer, reflecting off the terminating hard wall, and crossing back accumulates the attenuation

$$R = \exp\left[-\frac{2}{c} \int_0^L \sigma(x) dx\right] = \exp\left[-\frac{2\sigma_{\text{max}}L}{(m+1)c}\right], \quad (\text{B10})$$

so prescribing a design reflection  $R_0$  gives the Collino–Tsogka amplitude [9]

$$\sigma_{\text{max}} = -\frac{(m+1)c \ln R_0}{2L}, \quad (\text{B11})$$

used with the standard recipe  $m = 2$ ,  $R_0 = 10^{-3}$  [8]. The nominal  $R_0$  is realised only for layers of  $\gtrsim 8$  points; for thinner layers the discrete transition error dominates (a four-point layer reflects at the 0.5–2% level for grid-scale wavelengths), consistent with the saturation of Fig. 2(a).

When the CFS parameters are used they are graded with the same geometry,

$$\kappa(d) = 1 + (\kappa_{\text{max}} - 1)(d/L)^m, \quad \alpha(d) = \alpha_{\text{max}}(1 - d/L), \quad (\text{B12})$$

$\kappa$  rising from 1 at the interface to  $\kappa_{\text{max}}$  at the wall with the same polynomial as  $\sigma$ , and  $\alpha$  tapering linearly [8] from  $\alpha_{\text{max}}$  just inside the interface to zero at the wall (and vanishing identically outside the layer); in the implementation  $\alpha(d) = \alpha_{\text{max}}[1 - (\sigma(d)/\sigma_{\text{max}})^{1/m}]$ , which is identical. A nonzero  $\alpha$  improves grazing-incidence and evanescent absorption but unmatched the layer at low frequencies (the stretch is matched only for  $\omega \gg \alpha$ ), so the fixed-horizon absorption benchmarks of Sec. III B use  $\alpha_{\text{max}} = 0$ . Finally, the memory rescaling defaults to  $\gamma = \sqrt{2\sigma_{\text{max}}/h}$ , the choice that balances the injection and memory-source contributions to  $\lambda_{\text{max}}(H_1)$  (Appendix A).

### Appendix C: Schrödingerisation: discrete formulas and recovery

This appendix records the complete discrete recipe behind the summary of Sec. II C, with the exact sign, grid, and ordering conventions of our implementation; these are the conventions validated end to end by the demonstrations of Sec. IV C.

#### 1. Hermitian split and warped transform

The semi-discrete absorbing dynamics  $\dot{z} = Az$  is split as

$$A = H_1 + iH_2, \quad H_1 = \frac{A + A^\dagger}{2}, \quad H_2 = \frac{A - A^\dagger}{2i}, \quad (\text{C1})$$

with the dissipative part  $H_1$  and the Hamiltonian part  $H_2$  both Hermitian. The warped phase-space transform of Ref. [11] appends one auxiliary variable  $p$  and sets  $w(t, p) = e^{-p} z(t)$  for  $p \geq 0$  (extended to  $p < 0$  by the profile  $g$  of Sec. C3). The sign structure follows in one line: along the exact solution,  $\partial_t w = e^{-p} A z = A w$  and  $\partial_p w = -e^{-p} z = -w$ , so  $H_1 w = -H_1 \partial_p w$  and

$$\partial_t w = -H_1 \partial_p w + i H_2 w, \quad (\text{C2})$$

a transport equation in  $p$ : the non-unitary part of  $A$  has become advection of the auxiliary variable.

## 2. Discrete setup and per-mode evolution

The  $p$  register holds  $N_p = 2^{n_p}$  grid points on the periodic domain  $[-p_{\max}, p_{\max})$ ,

$$p_j = -p_{\max} + j \Delta p, \quad \Delta p = \frac{2p_{\max}}{N_p}, \quad (\text{C3})$$

$j = 0, \dots, N_p - 1$ , with dual frequencies  $\eta_k = \pi k / p_{\max}$  for  $k = -N_p/2, \dots, N_p/2 - 1$  in FFT ordering [i.e.  $\eta = 2\pi \times \text{fftfreq}(N_p, \Delta p)$ ]. We fix the Fourier *synthesis* convention  $w(\cdot, p) = \sum_k \hat{w}_k e^{+i\eta_k p}$  (NumPy:  $w = \text{ifft}(\hat{w})$  along the  $p$  axis,  $\hat{w} = \text{fft}(w)$ ). Substituting  $\partial_p \rightarrow i\eta_k$  in Eq. (C2) decouples the modes into the unitary evolutions of Eq. (5),

$$\begin{aligned} \frac{d\hat{w}_k}{dt} &= -i(\eta_k H_1 - H_2) \hat{w}_k \\ \implies \hat{w}_k(T) &= e^{-iT(\eta_k H_1 - H_2)} \hat{w}_k(0). \end{aligned} \quad (\text{C4})$$

Stacking the modes, the whole evolution is one Hamiltonian simulation,

$$H_{\text{tot}} = D_\eta \otimes H_1 - I_p \otimes H_2, \quad D_\eta = \text{diag}(\eta_k), \quad (\text{C5})$$

acting on the tensor order ( $p$  register, system register) with the  $p$  register most significant: on the circuit, qubits  $0, \dots, n_s - 1$  carry the system register (little-endian) and qubits  $n_s, \dots, n_s + n_p - 1$  the  $p$  register, so the flattened statevector index is  $j 2^{n_s} + s$ . The opening QFT<sup>†</sup> on the  $p$  register maps the grid amplitudes to  $\hat{w}/\sqrt{N_p}$  (the unitarily normalised FFT),  $e^{-iTH_{\text{tot}}}$  acts in the  $\eta$  basis, and the closing QFT returns to the  $p$  grid [Fig. 5(b)].

## 3. Initial data and the $C^1$ profile

The initial condition  $w(0, p_j) = g(p_j) z(0)$  is a product state across the two registers,

$$|W(0)\rangle = \frac{|g\rangle \otimes |z_0\rangle}{N_0}, \quad N_0 = \|g\|_2 \|z_0\|_2, \quad (\text{C6})$$

with the normalisation  $N_0$  recorded classically and undone at recovery. The plain profile  $g(p) = e^{-|p|}$  is kinked

at  $p = 0$  and limits the  $p$  discretisation to  $O(\Delta p)$ ; the  $C^1$  cubic of Ref. [13, Remark 4.9] replaces it on  $(-1, 0)$ ,

$$g(p) = \begin{cases} e^{-p}, & p \geq 0, \\ \left(\frac{3}{e} - 3\right)p^3 + \left(\frac{4}{e} - 5\right)p^2 - p + 1, & -1 < p < 0, \\ e^p, & p \leq -1, \end{cases} \quad (\text{C7})$$

restoring  $O(\Delta p^2)$ . The junctions are  $C^1$  by construction: at  $p = 0^-$  the cubic gives  $g = 1, g' = -1$ , matching  $e^{-p}$ ; at  $p = -1^+$  it gives  $g = g' = e^{-1}$ , matching  $e^p$ . Because  $g(p) = e^{-p}$  *exactly* for  $p \geq 0$ , the recovery formula below is unchanged by the smoothing. Measured on the 1D CPML benchmark of Fig. 2(c) ( $2^5$  spatial points,  $n_{\text{pml}} = 8$ ,  $\sigma_{\max} = 1$ ,  $T = 30$ ,  $p_{\max} = 18$ ): the recovery error at the default slice falls from  $4.1 \times 10^{-2}$  to  $6.3 \times 10^{-4}$  over  $n_p = 6-11$  for the kinked profile and from  $1.3 \times 10^{-2}$  to  $2.4 \times 10^{-6}$  for the cubic, with fitted slopes of  $\log_2(\text{error})$  versus  $n_p$  of  $-1.2$  and  $-2.6$  ( $-1.6$  and  $-3.4$  on the pre-floor points  $n_p \leq 9$ )—at or above the nominal first- and second-order rates, confirming the one-order upgrade of Ref. [13] in this setting.

## 4. Recovery: threshold, wraparound, plateau

The solution is recovered from a single grid slice,

$$z(T) = e^{p^*} w(T, p^*), \quad p^* \geq \lambda_{\max}^+(H_1) T, \quad (\text{C8})$$

where  $\lambda_{\max}^+ = \max\{0, \lambda_{\max}(H_1)\}$  and the validity threshold on  $p^*$  is Theorem 3.1 of Ref. [13]. When  $H_1 \preceq 0$  (the collapsed 1D CPML and the sponge) any  $p^* > 0$  is admissible, and the implementation defaults to the slice three grid points above  $p = 0$ , i.e.  $p^* = 3\Delta p$ . For indefinite  $H_1$  (the 2D CPML memory form of Lemma 1) the slice must be placed explicitly at  $p^* \geq \lambda^+ T$ ; below the threshold the recovered state is  $O(1)$  wrong (Fig. 3).

Two further discrete conditions guard the slice. *Wraparound*: in Eq. (C2) an  $H_1$ -eigencomponent with eigenvalue  $\lambda$  is advected in  $p$  with velocity  $\lambda$ , so the most strongly damped components travel a distance  $\lambda^- T$  *leftwards*, with  $\lambda^- = |\lambda_{\min}(H_1)|$  ( $= \sigma_{\max}$  for the diagonal absorbers). The discrete  $p$  domain is periodic, so  $p_{\max}$  must be wide enough that this leftward-transported mass does not wrap through  $p = -p_{\max}$  back into the recovery region. *Plateau*: since  $e^p w(T, p) = z(T)$  for every admissible  $p$ , the function  $p \mapsto e^p \|w(T, p)\|$  must be flat across the recovery window. The implementation tabulates it over the first twelve grid points right of  $p = 0$  and reports the relative peak-to-peak spread; a flat plateau certifies that  $p_{\max}$ ,  $N_p$ , and the profile are adequate (our regression tests require a spread below  $5 \times 10^{-2}$ ), while a tilted or stepped plateau flags wraparound or an under-resolved profile.

## 5. Measurement interpretation

The pipeline is unitary, so the final statevector is exactly  $W(T)/N_0$ , holding  $w_s(T, p_j)/N_0$  at index  $j 2^{n_s} + s$ . Measuring the  $p$  register returns outcome  $p_j$  with probability  $\|w(T, p_j)\|^2/N_0^2$  and collapses the system register onto  $w(T, p_j)/\|w(T, p_j)\| = z(T)/\|z(T)\|$ : every slice in the admissible window yields the *same* normalised state, so post-selection may accept the whole plateau window rather than a single outcome, summing the corresponding probabilities. The success probability of the slice at  $p^*$  is, up to  $p$ -discretisation error,

$$\Pr(p^*) = \frac{e^{-2p^*} \|z(T)\|^2}{N_0^2} \propto e^{-2p^*}, \quad (\text{C9})$$

which at the threshold slice  $p^* = \lambda^+ T$  is the  $e^{-2\lambda^+ T}$  post-selection penalty of Sec. III C—the cost that the Lyapunov symmetrizer of Sec. III D replaces by the  $T$ -independent factor  $\kappa_2^2(S)$ . On the statevector simulator used for the demonstrations of Sec. IV C, the slice is read off rather than sampled: the final statevector is reshaped to an  $N_p \times 2^{n_s}$  array, row  $j$  at  $p^*$  is extracted, and the result is rescaled by  $e^{p_j} N_0$ . This is deterministic and exact, so the reported errors measure the Trotter and discretisation error only; on hardware the same slice would additionally cost the factor  $e^{-2p^*}$  of Eq. (C9) in repetitions.

### Appendix D: Validation protocol and convergence data

This appendix records the classical validation protocol behind every reflection number quoted in Sec. III B and the raw convergence data behind the fitted orders of Sec. IV B [Figs. 2(a–d)].

#### 1. Gold-standard reflection protocol

Measuring the residual reflection of an absorbing layer against the absorbing dynamics itself is circular; the gold standard is the open-domain solution. The truncated domain has  $N = 128$  points (seven grid qubits) with the CPML occupying  $n_{\text{pml}}$  points at each end. The initial data are a Gaussian velocity bump centred at  $j = 64$  with width 3 cells and  $w(0) = 0$ , normalised to  $\|z_0\|_2 = 1$ . The reference embeds the *same* truncated data centred in a  $4\times$  hard-wall domain ( $N_{\text{ref}} = 512$ , embedding offset  $\Delta = (N_{\text{ref}} - N)/2 = 192$ ) and evolves it with the closed-domain unitary  $e^{-iH_{\text{ref}}T}$ . Both evolutions are computed by dense matrix exponentials, so the comparison contains no time-integration error. The reported error is the pointwise  $\ell_2$  difference on the interior window,

$$E(T)^2 = \sum_{f \in \{v, w\}} \sum_{j=n_{\text{pml}}}^{N-n_{\text{pml}}-1} |f_j(T) - f_{j+\Delta}^{\text{ref}}(T)|^2. \quad (\text{D1})$$

TABLE V. Gold-standard true reflection (1D CPML,  $N = 128$  versus the  $N = 512$  hard-wall reference,  $m = 2$ ): plateau error  $\max_{T \in \{100, 120, 140\}} E(T)$  and relative plateau spread. Upper block: layer-width sweep at  $R_0 = 10^{-3}$ ; lower block: design-reflection sweep at  $n_{\text{pml}} = 12$ .

| $n_{\text{pml}}$ | $R_0$     | plateau error        | plateau spread        |
|------------------|-----------|----------------------|-----------------------|
| 4                | $10^{-3}$ | $6.7 \times 10^{-4}$ | $1.0 \times 10^{-9}$  |
| 6                | $10^{-3}$ | $2.6 \times 10^{-4}$ | $3.3 \times 10^{-10}$ |
| 8                | $10^{-3}$ | $2.6 \times 10^{-4}$ | $2.2 \times 10^{-10}$ |
| 12               | $10^{-3}$ | $3.4 \times 10^{-4}$ | $1.7 \times 10^{-9}$  |
| 16               | $10^{-3}$ | $4.2 \times 10^{-4}$ | $3.3 \times 10^{-5}$  |
| 12               | $10^{-2}$ | $4.5 \times 10^{-3}$ | $1.5 \times 10^{-10}$ |
| 12               | $10^{-3}$ | $3.4 \times 10^{-4}$ | $1.7 \times 10^{-9}$  |
| 12               | $10^{-4}$ | $4.3 \times 10^{-5}$ | $1.9 \times 10^{-7}$  |

Two checks make the protocol sound. (i) *Window alignment*: at  $T = 0$  the windowed difference in Eq. (D1) evaluates to exactly zero in floating point, pinning the embedding offset and the index conventions. (ii) *Causal validity*: with  $c = 1$ ,  $h = 1$  the pulse meets the layer interface at depth  $j = n_{\text{pml}}$  and its reflection is back at the centre by  $T \approx 2(64 - n_{\text{pml}}) \leq 120$  for  $n_{\text{pml}} \geq 4$ , whereas the far walls of the reference domain are first felt inside the comparison window only at  $T \approx 380$ . The plateau times  $T \in \{100, 120, 140\}$  therefore capture the complete round-trip reflection of the truncated domain while the reference is still uncontaminated. The reported reflection is the maximum of  $E(T)$  over the three plateau times, and the relative plateau spread  $[(\max - \min)/\text{mean}]$  is the flatness diagnostic. A reduced instance of the same protocol ( $N = 64$  versus 256,  $T = 20$  and 60, asserted  $< 10^{-3}$ ) is pinned in the unit-test suite (Sec. D 6).

#### 2. Reflection versus layer width and design reflection

Table V interprets as follows. The curve is the sum of two competing contributions. At the thinnest layer ( $n_{\text{pml}} = 4$ ) the discrete transition error dominates: the measured reflection sits  $9.4\times$  above the discrete wall-return prediction. For  $n_{\text{pml}} \geq 8$  the transition error is negligible and the reflection is set by the wall return, which *rises toward the design value from below* as the layer widens: node sampling of the polynomial profile overestimates  $\int \sigma dx$  by a factor  $1 + O(1/n_{\text{pml}})$  (exactly  $\Sigma \sigma h / \int \sigma = 1.38, 1.25, 1.19, 1.13, 1.09$  for  $n_{\text{pml}} = 4\text{--}16$ ), so the effective round-trip reflection is  $R_{\text{eff}} = R_0^{\Sigma \sigma h / \int \sigma} = 7.1 \times 10^{-5}, 1.7 \times 10^{-4}, 2.7 \times 10^{-4}, 4.2 \times 10^{-4}, 5.2 \times 10^{-4}$  across the sweep; the measured values track this prediction within 5–20% for  $n_{\text{pml}} \geq 8$ . Thin layers therefore *over-perform* the nominal  $R_0$  (at the price of larger transition error), and widening the layer converges the reflection up to the design value. We verified this directly

in the SEISMIC\_CPML reference implementation [8] (the 2D isotropic elastic code, unmodified kernels, same gold-standard protocol,  $\alpha = 0$ ,  $K = 1$ ): its  $P$ -wave reflection is likewise non-monotonic and rises from  $n_{\text{pml}} = 12$  to 16 ( $5.8 \times 10^{-4} \rightarrow 6.5 \times 10^{-4}$ , against a predicted  $6.5 \times 10^{-4} \rightarrow 7.2 \times 10^{-4}$ ), converging up toward the design  $R_0$  and matching its own discrete-sum prediction within  $\sim 10\%$  for  $n_{\text{pml}} \geq 12$ . Two scheme-dependent differences: the staggered elastic scheme samples the profile at both full and half points, so its quadrature ratio is  $1 + 3/(4n_{\text{pml}})$  rather than our  $\sim 1 + 3/(2n_{\text{pml}})$ ; and the minimum sits at  $n_{\text{pml}} \approx 12$  there because  $S$ -wave transition reflection (whose design value  $R_0^{c_p/c_s}$  is far smaller) and the coarser wavelength resolution keep the transition term dominant further out. The effect is invisible in normal use of that code, where the layer width is fixed at  $\sim 10$  cells.

The miscalibration admits a one-line remedy, which we provide as an option (`calibration='discrete'`): rescale the staggered profile *pair* by a single common factor so that the characteristic damping sum  $\Sigma(\sigma_v + \sigma_w)h/2$  equals the design integral exactly. The realised reflection is then flat at the design value, 0.81–1.08  $R_0$  across the whole sweep  $n_{\text{pml}} = 4$ –16 [red squares in Fig. 2(a)], i.e. the Collino–Tsogka formula becomes exact on the grid. The common factor is essential: rescaling each profile *separately* gives  $\sigma_v$  and  $\sigma_w$  different damping functions (the node and midpoint quadratures differ), an in-layer impedance mismatch that reflects at  $O(1/n_{\text{pml}})$ —measured at  $1.2 \times 10^{-2}$  to  $6.9 \times 10^{-2}$ , i.e. ten to seventy times *worse* than no calibration at all. This remedy applies verbatim to classical CPML codes. That the level is  $R_0$ -controlled rather than an absolute discretisation floor is established by the lower block: at  $n_{\text{pml}} = 12$  the measured reflection tracks the design value over two decades, with ratios  $E/R_0 = 0.45, 0.34, 0.43$ —the Collino–Tsogka formula is predictive within a factor  $\sim 2.3$  in this discretisation, and lowering  $R_0$  lowers the true reflection commensurately, down to  $4.3 \times 10^{-5}$  at  $R_0 = 10^{-4}$ . This is the floor behaviour reported for classical CPML implementations by Komatitsch and Martin [8]. The plateau is flat to  $\leq 2 \times 10^{-9}$  (relative) in six of the eight rows; even the two most weakly damped rows ( $n_{\text{pml}} = 16$  and  $R_0 = 10^{-4}$ ) vary by only  $3.3 \times 10^{-5}$  and  $1.9 \times 10^{-7}$  of their mean across  $T \in \{100, 120, 140\}$ , i.e. the round-trip transient has fully decayed and the quoted numbers are converged in  $T$ .

### 3. Recovery order in the $p$ register

Table VI gives the recovery error of the full Schrödingerised pipeline (exact per-mode evolution, no Trotter) as a function of the  $p$ -register size, for the two warping profiles of Sec. II C. The errors are absolute per unit initial norm; by  $T = 30$  the layer has absorbed all but  $\|e^{AT}z_0\|_2 = 1.4 \times 10^{-3}$  of the state, so resolving the residual field demands the higher-order profile. Both pro-

TABLE VI. Schrödingerisation recovery error versus  $p$ -register size  $n_p$  for the kinked  $e^{-|p|}$  profile and the  $C^1$  cubic profile of Ref. [13] [1D CPML benchmark,  $N = 32$ ,  $n_{\text{pml}} = 8$ ,  $\sigma_{\text{max}} = 1$ ,  $T = 30$ ,  $p \in [-18, 18]$ ,  $\Delta p = 36/2^{n_p}$ ]. Errors are  $\|z_{\text{rec}} - e^{AT}z_0\|_2$  with  $\|z_0\|_2 = 1$ , recovered at the library-default slice  $p^* = 3\Delta p$  (valid at any  $p^* > 0$  since  $H_1 \leq 0$  here). The fitted orders are least-squares slopes of  $\log_2 E$  versus  $n_p$ .

| $n_p$                               | $\Delta p$ | $e^{- p }$ profile   | $C^1$ cubic profile  |
|-------------------------------------|------------|----------------------|----------------------|
| 6                                   | 0.563      | $4.1 \times 10^{-2}$ | $1.3 \times 10^{-2}$ |
| 7                                   | 0.281      | $1.6 \times 10^{-2}$ | $2.6 \times 10^{-3}$ |
| 8                                   | 0.141      | $3.6 \times 10^{-3}$ | $1.7 \times 10^{-4}$ |
| 9                                   | 0.0703     | $1.5 \times 10^{-3}$ | $1.3 \times 10^{-5}$ |
| 10                                  | 0.0352     | $1.1 \times 10^{-3}$ | $5.9 \times 10^{-6}$ |
| 11                                  | 0.0176     | $6.3 \times 10^{-4}$ | $2.4 \times 10^{-6}$ |
| order in $\Delta p$ (all points)    |            | 1.23                 | 2.64                 |
| order in $\Delta p$ ( $n_p = 6$ –9) |            | 1.64                 | 3.40                 |

TABLE VII. Gate-level Trotter error versus step count [1D CPML,  $N = 8$ ,  $n_{\text{pml}} = 2$ ,  $\sigma_{\text{max}} = 1$ ,  $T = 4$ ,  $n_p = 5$ ,  $p_{\text{max}} = 8$ ; transpiled circuits on a statevector simulator; errors  $\|z_{\text{circ}} - z_{\text{ref}}\|_2$  with  $\|z_0\|_2 = 1$ ]. “vs exact S.” uses the exact Schrödingerised evolution (exact per-mode propagation at the same  $p$  discretisation) as reference, isolating the splitting error; “vs  $e^{AT}z_0$ ” uses the full matrix-exponential reference and therefore contains the  $p$ -discretisation floor, measured independently as  $1.29 \times 10^{-2}$ . Order 2 symmetrises both Trotter levels (Strang outer split, palindromic term order inner).

| steps | order 1               |                       | order 2               |                       |
|-------|-----------------------|-----------------------|-----------------------|-----------------------|
|       | vs exact S.           | vs $e^{AT}z_0$        | vs exact S.           | vs $e^{AT}z_0$        |
| 10    | $1.36 \times 10^{-1}$ | $1.32 \times 10^{-1}$ | $3.24 \times 10^{-2}$ | $3.40 \times 10^{-2}$ |
| 20    | $6.55 \times 10^{-2}$ | $6.27 \times 10^{-2}$ | $8.02 \times 10^{-3}$ | $1.46 \times 10^{-2}$ |
| 40    | $3.25 \times 10^{-2}$ | $3.11 \times 10^{-2}$ | $2.00 \times 10^{-3}$ | $1.29 \times 10^{-2}$ |
| 80    | $1.62 \times 10^{-2}$ | $1.75 \times 10^{-2}$ | $4.99 \times 10^{-4}$ | $1.29 \times 10^{-2}$ |
| slope | -1.02                 | -0.98                 | -2.01                 | -0.44 <sup>a</sup>    |

<sup>a</sup> Floor-limited: not a meaningful order, see text.

files meet or exceed their nominal orders [ $O(\Delta p)$  kinked,  $O(\Delta p^2)$  cubic, Remark 4.9 of Ref. [13]]: the full-sweep fits give orders 1.23 and 2.64, and the pre-floor fits ( $n_p = 6$ –9) give 1.64 and 3.40. Because the default slice  $p^* = 3\Delta p$  moves with refinement, we re-evaluated both series at the fixed slice  $p^* = 1$ : the fitted orders become 1.64 and 3.25, so the profile ordering is not a slice artefact. At  $n_p = 11$  the cubic profile is  $\sim 260\times$  more accurate ( $2.4 \times 10^{-6}$  versus  $6.3 \times 10^{-4}$ ); the flattening of its last two points marks the approach to the recovery floor of this configuration.

TABLE VIII. Stagger ablation under the gold-standard protocol (Sec. D 1; 1D CPML,  $n_{\text{pml}} = 12$ ,  $R_0 = 10^{-3}$ ): plateau reflection versus the stagger offset  $s$  of the  $\sigma_w$  sampling,  $x_j = (j + s)h$ .

| $\sigma_w$ stagger $s$     | plateau error        | degradation   |
|----------------------------|----------------------|---------------|
| $-1/2$ (correct half-cell) | $3.4 \times 10^{-4}$ | 1             |
| 0 (unstaggered)            | $4.1 \times 10^{-2}$ | $\approx 120$ |
| $+1/2$ (wrong sign)        | $8.1 \times 10^{-2}$ | $\approx 235$ |

#### 4. Trotter orders at gate level

Table VII separates the two error sources of the circuit pipeline of Sec. IV B. Against the exact Schrödingerised evolution the splitting error is clean: fitted slope  $-1.02$  at order 1 and  $-2.01$  at order 2 with both Trotter levels symmetrised [Fig. 2(d)]. Against the full  $e^{AT}z_0$  reference the order-2 series saturates at  $1.29 \times 10^{-2}$  from 40 steps onward—exactly the independently measured  $p$ -discretisation floor of this deliberately coarse  $n_p = 5$  register—so the nominal slope  $-0.44$  in that column is floor-limited rather than a defect of the splitting. The order-1 series does not saturate within the sweep because its splitting error stays above the floor. Enlarging the  $p$  register lowers the floor at the rates of Table VI.

#### 5. Stagger ablation

The damping profiles are sampled at  $x_j = (j + s)h$  with the stagger  $s$  matched to where each field lives (Sec. II B): for the forward-difference convention,  $v$  at integer nodes ( $s = 0$ ) and  $w$  at the half-cells  $x_{j-1/2}$  ( $s = -1/2$ ). Rerunning the gold-standard protocol ( $n_{\text{pml}} = 12$ ,  $R_0 = 10^{-3}$ ) with the  $\sigma_w$  stagger ablated gives Table VIII: misplacing the profile by half a cell costs two orders of magnitude in true reflection. The stagger is load-bearing, not cosmetic.

#### 6. Test suite

The classical and circuit layers are pinned by 24 unit tests (`tests/test_absorbing.py`), grouped by the property classes they assert. (i) *Generator structure and spectra*: the Collino–Tsogka profile amplitude and support;  $\max \text{Re eig}(A) \leq 0$  to round-off for the collapsed 1D, general CFS ( $\kappa_{\text{max}} = 2$ ,  $\alpha_{\text{max}} = 0.05$ ), and 2D memory-form generators; the Hermitian split  $A = H_1 + iH_2$  with  $H_1 = -\text{diag}(\sigma) \preceq 0$  for the collapsed 1D form and the sponge. (ii) *Collapse*: the 1D memory form reproduces the collapsed diagonal damping to  $10^{-10}$ . (iii) *Absorption and reflection bounds*: interior energy below  $10^{-5}$  where the hard-wall reference retains more than 0.5; round-trip amplitude reflection below  $5 \times 10^{-3}$ ; 2D CPML and sponge energy decay; and the gold-standard true reflection

below  $10^{-3}$  on the reduced instance. (iv) *Recovery thresholds*: scalar decay recovered to  $10^{-4}$ ; the cubic profile beating the kinked profile at equal resolution; 1D recovery below  $2 \times 10^{-3}$  with a plateau-flatness diagnostic; 2D sponge recovery below  $10^{-3}$  with  $\lambda^+ = 0$  asserted; and the 2D CPML threshold of Sec. III C verified in both directions—accurate above  $p^* = \lambda^+T + 1$  and at least ten times worse below. (v) *Circuit-versus-matrix agreement*: the exact-Hamiltonian-gate path against the numpy per-mode pipeline to  $10^{-8}$  (pinning the QFT/FFT conventions and qubit ordering); the Trotterised Bell circuit to  $5 \times 10^{-2}$ ; the separated 2D operator strings against the block Hamiltonian to  $10^{-13}$ ; the per-step orders  $O(\delta t^2)$  and  $O(\delta t^3)$  of the projector-aware Bell circuit and its Strang variant; and the 17-qubit end-to-end gate-level run to  $3 \times 10^{-2}$ . (vi) *Guards and regression*: rejection of out-of-range recovery slices, degenerate layers, and inconsistent damping vectors; and a regression test pinning the untouched closed-domain library (the mixed Dirichlet/Neumann Laplacian stencil). Every bound above is asserted numerically, so any change that degrades a spectrum, a reflection, a recovery threshold, or a circuit convention fails the suite.

### Appendix E: The Lyapunov symmetrizer: construction, exact properties, and compressibility data

This appendix collects the classical construction behind Sec. IIID, the exact identity underlying Eq. (13) with its proof, the complete conditioning data of which Table II is an excerpt, the grid-size saturation measurements, and the compressibility study of the symmetrizer  $S$ .

#### 1. Construction and classical cost

The absorbing generator  $A$  is marginally stable—we measure  $\max \text{Re eig}(A) = 0$  to round-off (Sec. IIIB)—so the Lyapunov equation for  $A$  itself is singular. We therefore shift,  $A_\epsilon = A - \epsilon I$ , which is Hurwitz with margin  $\epsilon$ , and solve

$$A_\epsilon^\dagger W + W A_\epsilon = -I, \quad W = \int_0^\infty e^{tA_\epsilon^\dagger} e^{tA_\epsilon} dt \succ 0, \quad (\text{E1})$$

the integral being the unique Hermitian solution. The symmetrizer is the Hermitian square root  $S = W^{1/2}$ , with conditioning  $\kappa_2(S) = \sqrt{\lambda_{\text{max}}(W)/\lambda_{\text{min}}(W)}$ . The shift is undone classically by the scalar factor  $e^{\epsilon T}$  [Eq. (12)].

Numerically, Eq. (E1) is solved by the Bartels–Stewart algorithm (one complex Schur decomposition of  $A_\epsilon^\dagger$  plus a triangular substitution; `scipy.linalg.solve_continuous_lyapunov`), and  $S$ ,  $S^{-1}$  follow from one Hermitian eigendecomposition of

$W$ —three dense  $O(n^3)$  kernels in total. Measured wall times at the largest completed size,  $n = 4096$  ( $32 \times 16$  grid,  $\sigma_{\max} = 0.5$ ): the Lyapunov solve takes 268 s at  $\epsilon = 10^{-2}$  and 244 s at  $\epsilon = 10^{-3}$ , the eigendecomposition of  $W$  takes 31 s, the residual verification 2.5–2.6 s, and the sparse Lanczos evaluation of  $\lambda^+ = \lambda_{\max}^+(H_1)$  is negligible (0.05 s). The  $n = 8192$  point was gated off by cubic extrapolation of the measured  $n = 4096$  time (projected 40.3 min per  $\epsilon$ , over the 25-min budget) and by memory (a dense complex  $8192^2$  matrix occupies 1 GiB and the solve holds six to eight of them, against 5.1 GiB free). Across all 18 configurations of Table IX the relative Lyapunov residual  $\|A_\epsilon^\dagger W + W A_\epsilon + I\|_F / \|W\|_F$  is at most  $1.4 \times 10^{-14}$ .

## 2. An exact identity for the transformed Hermitian part

The equality sign in Eq. (13) is not an estimate. With the unit right-hand side  $-I$  in Eq. (E1), the Hermitian part of the transformed generator is an explicit function of  $W$ :

**Proposition 1** *Let  $A \in \mathbb{C}^{n \times n}$  have spectral abscissa at most zero, fix  $\epsilon > 0$ , set  $A_\epsilon = A - \epsilon I$ , let  $W \succ 0$  solve  $A_\epsilon^\dagger W + W A_\epsilon = -I$ , and let  $S = W^{1/2}$  be the Hermitian positive-definite square root. Then, exactly,*

$$\frac{1}{2}(S A_\epsilon S^{-1} + S^{-1} A_\epsilon^\dagger S) = -\frac{1}{2}W^{-1} \prec 0, \quad (\text{E2})$$

$$\frac{1}{2}(S A S^{-1} + S^{-1} A^\dagger S) = \epsilon I - \frac{1}{2}W^{-1}. \quad (\text{E3})$$

In particular the  $\epsilon$ -shifted transform is strictly dissipative with  $\lambda_{\max}(\tilde{H}_1^{(\epsilon)}) = -1/(2\lambda_{\max}(W))$ , and the transform of the original  $A$  has  $\lambda_{\max}(\tilde{H}_1) = \epsilon - 1/(2\lambda_{\max}(W)) < \epsilon$ .

*Proof.* Since  $A_\epsilon$  is Hurwitz, the integral in Eq. (E1) converges and is the unique Hermitian solution; it is positive definite, so  $S = W^{1/2}$  exists, is Hermitian, and  $S^2 = W$ . Write  $\tilde{A} = S A_\epsilon S^{-1}$  and take any  $x \in \mathbb{C}^n$  with  $x = S y$ . By congruence,

$$x^\dagger (\tilde{A} + \tilde{A}^\dagger) x = y^\dagger (W A_\epsilon + A_\epsilon^\dagger W) y = -\|y\|^2 = -x^\dagger W^{-1} x, \quad (\text{E4})$$

using  $S \tilde{A} S = W A_\epsilon$  and  $S \tilde{A}^\dagger S = A_\epsilon^\dagger W$ . Two Hermitian matrices with equal quadratic forms are equal, so  $\tilde{A} + \tilde{A}^\dagger = -W^{-1}$ , which is Eq. (E2); its largest eigenvalue is  $-\frac{1}{2}\lambda_{\min}(W^{-1}) = -1/(2\lambda_{\max}(W))$ . For Eq. (E3),  $S A S^{-1} = S A_\epsilon S^{-1} + \epsilon I$  and  $\epsilon I$  is Hermitian, so the Hermitian part shifts by  $\epsilon I$  and the eigenvalues are  $\epsilon - 1/(2\lambda_j(W))$ . ■

The identity is verified in every measured configuration: in all 18 rows of Table IX the directly computed  $\lambda_{\max}(\tilde{H}_1)$  (dense eigendecomposition of the Hermitian part of  $S A S^{-1}$ ) agrees with the prediction  $\epsilon - 1/(2\lambda_{\max}(W))$  to a relative deviation of at most  $3.8 \times 10^{-12}$ —far beyond the four-digit agreement quoted in the

main text—and the shifted column equals the unshifted one minus  $\epsilon$  to the same precision. A direct check of the matrix identity (E2) on an  $n = 50$  instance gives a Frobenius residual of  $8.9 \times 10^{-14}$ . Proposition 1 also explains the choice made in Sec. IIID: the transform of the *original*  $A$  is not dissipative ( $\lambda_{\max}(\tilde{H}_1) \approx \epsilon > 0$ ), whereas the transform of  $A_\epsilon$  is, strictly; we therefore evolve the shifted system and undo  $e^{\epsilon T}$  classically.

## 3. The full conditioning table

Table IX lists all 18 measured configurations (three grids, two damping strengths, three shifts). Three regularities are worth recording. (i)  $\lambda_{\min}(W)$  is essentially independent of  $\epsilon$  (0.66–0.68 at  $\sigma_{\max} = 0.5$ , 0.37 at  $\sigma_{\max} = 1.0$ ), so the entire  $\epsilon$  dependence of  $\kappa_2(S)$  is carried by  $\lambda_{\max}(W)$ , consistent with Proposition 1, in which  $\lambda_{\max}(W)$  is pinned to the inverse dissipativity margin of the shifted transform. (ii) Least-squares fits of  $\ln \kappa_2(S)$  against  $\ln \epsilon$  per (grid,  $\sigma_{\max}$ ) configuration give slopes between  $-0.64$  and  $-1.03$  (the range quoted as  $\epsilon^{-0.6}$  to  $-1.0$  in Sec. IIID), steepening with grid size at fixed  $\epsilon$  range. (iii) The crossover times  $T^* = \ln \kappa^2 / (2\lambda^+)$  span 6.7–16.0 and grow only logarithmically with  $\kappa_2(S)$ . For comparison, the eigenvector symmetrizer  $W = (V V^\dagger)^{-1}$  yields  $\kappa_2(S) = 2.9 \times 10^8$ – $6.9 \times 10^8$  where it exists and fails outright ( $V V^\dagger$  numerically singular) on both  $16 \times 8$  configurations; the Lyapunov construction is four to six orders of magnitude better conditioned and never fails.

## 4. Saturation of $\kappa_2(S)$ with grid size

Table X extends the  $\sigma_{\max} = 0.5$  series to  $n = 4096$ . At  $\epsilon = 10^{-2}$  the conditioning is flat:  $\kappa_2(S)$  decreases from 566.9 to 564.5 between  $n = 2048$  and  $n = 4096$  (local log-log slope  $-0.006$ ). At  $\epsilon = 10^{-3}$  it still rises but visibly flattens. Power-law fits confirm that a power law is the wrong model: at  $\epsilon = 10^{-2}$  the fitted exponent drops from 0.25 ( $r^2 = 0.963$ ,  $n = 512$ – $2048$ ) to 0.16 ( $r^2 = 0.823$ ,  $n = 512$ – $4096$ ), and at  $\epsilon = 10^{-3}$  from 0.86 ( $r^2 = 0.943$ ) to 0.62 ( $r^2 = 0.875$ )—falling exponents with degrading fit quality, i.e.  $\kappa$  is saturating, not following a power law. The drivers saturate individually:  $\lambda^+$  grows from 0.6829 to 0.6940 over the eightfold range in  $n$  and is then identical to ten digits between  $n = 2048$  and  $n = 4096$ , and  $\lambda_{\min}(W)$  stays at 0.655–0.681 throughout. The conditioning is therefore  $\epsilon$ -controlled, not grid-controlled: mesh refinement does not inflate the symmetrizer cost.

## 5. Compressibility of $S$

*a. Protocol.* At  $\epsilon = 10^{-2}$ ,  $\sigma_{\max} = 0.5$ , we measured the structure of  $S$  on the  $n = 512$  grid and confirmed on the  $n = 1024$  grid, and scored

TABLE IX. Full Lyapunov-symmetrizer data set; Table II of the main text is the excerpt obtained by dropping the  $\lambda_{\min}(W)$  and unshifted-transform columns. Grids are listed in points ( $n$  is the full state dimension including the eight blocks).  $\lambda^+ = \lambda_{\max}^+(H_1)$  is the indefiniteness of the untransformed generator;  $\kappa_2(S) = \sqrt{\text{cond}_2(W)}$ ;  $\lambda_{\max}(\tilde{H}_1)$  is the largest Hermitian-part eigenvalue of the transform  $SAS^{-1}$  of the original generator and  $\lambda_{\max}(\tilde{H}_1^{(\epsilon)})$  that of the  $\epsilon$ -shifted transform  $SA_\epsilon S^{-1}$ ;  $T^* = \ln \kappa^2 / (2\lambda^+)$  is the crossover time. Every row satisfies the identities of Proposition 1:  $\lambda_{\max}(\tilde{H}_1)$  equals  $\epsilon - 1/(2\lambda_{\max}(W))$  to a relative deviation  $\leq 3.8 \times 10^{-12}$ , and  $\lambda_{\max}(\tilde{H}_1^{(\epsilon)}) = \lambda_{\max}(\tilde{H}_1) - \epsilon$ . Relative Lyapunov residuals are  $\leq 1.4 \times 10^{-14}$ .

| Grid           | $n$  | $\sigma_{\max}$ | $\epsilon$ | $\lambda^+(H_1)$ | $\lambda_{\min}(W)$ | $\kappa_2(S)$       | $\lambda_{\max}(\tilde{H}_1)$ | $\lambda_{\max}(\tilde{H}_1^{(\epsilon)})$ | $T^*$ |
|----------------|------|-----------------|------------|------------------|---------------------|---------------------|-------------------------------|--------------------------------------------|-------|
| $8 \times 8$   | 512  | 0.5             | $10^{-2}$  | 0.683            | 0.669               | 402.1               | $9.9954 \times 10^{-3}$       | $-4.63 \times 10^{-6}$                     | 8.8   |
| $8 \times 8$   | 512  | 0.5             | $10^{-3}$  | 0.683            | 0.681               | 3009                | $9.9992 \times 10^{-4}$       | $-8.10 \times 10^{-8}$                     | 11.7  |
| $8 \times 8$   | 512  | 0.5             | $10^{-4}$  | 0.683            | 0.683               | $1.105 \times 10^4$ | $9.9994 \times 10^{-5}$       | $-6.00 \times 10^{-9}$                     | 13.6  |
| $8 \times 8$   | 512  | 1.0             | $10^{-2}$  | 0.927            | 0.373               | 509.3               | $9.9948 \times 10^{-3}$       | $-5.17 \times 10^{-6}$                     | 6.7   |
| $8 \times 8$   | 512  | 1.0             | $10^{-3}$  | 0.927            | 0.376               | 2851                | $9.9984 \times 10^{-4}$       | $-1.64 \times 10^{-7}$                     | 8.6   |
| $8 \times 8$   | 512  | 1.0             | $10^{-4}$  | 0.927            | 0.377               | 9762                | $9.9986 \times 10^{-5}$       | $-1.39 \times 10^{-8}$                     | 9.9   |
| $16 \times 8$  | 1024 | 0.5             | $10^{-2}$  | 0.687            | 0.661               | 506.3               | $9.9970 \times 10^{-3}$       | $-2.95 \times 10^{-6}$                     | 9.1   |
| $16 \times 8$  | 1024 | 0.5             | $10^{-3}$  | 0.687            | 0.673               | 7009                | $9.9998 \times 10^{-4}$       | $-1.51 \times 10^{-8}$                     | 12.9  |
| $16 \times 8$  | 1024 | 0.5             | $10^{-4}$  | 0.687            | 0.674               | $3.052 \times 10^4$ | $9.9999 \times 10^{-5}$       | $-7.96 \times 10^{-10}$                    | 15.0  |
| $16 \times 8$  | 1024 | 1.0             | $10^{-2}$  | 0.956            | 0.369               | 749.6               | $9.9976 \times 10^{-3}$       | $-2.41 \times 10^{-6}$                     | 6.9   |
| $16 \times 8$  | 1024 | 1.0             | $10^{-3}$  | 0.956            | 0.372               | 6981                | $9.9997 \times 10^{-4}$       | $-2.76 \times 10^{-8}$                     | 9.3   |
| $16 \times 8$  | 1024 | 1.0             | $10^{-4}$  | 0.956            | 0.373               | $2.732 \times 10^4$ | $9.9998 \times 10^{-5}$       | $-1.80 \times 10^{-9}$                     | 10.7  |
| $16 \times 16$ | 2048 | 0.5             | $10^{-2}$  | 0.694            | 0.655               | 566.9               | $9.9976 \times 10^{-3}$       | $-2.37 \times 10^{-6}$                     | 9.1   |
| $16 \times 16$ | 2048 | 0.5             | $10^{-3}$  | 0.694            | 0.667               | 9856                | $9.9999 \times 10^{-4}$       | $-7.71 \times 10^{-9}$                     | 13.3  |
| $16 \times 16$ | 2048 | 0.5             | $10^{-4}$  | 0.694            | 0.669               | $6.612 \times 10^4$ | $1.0000 \times 10^{-4}$       | $-1.71 \times 10^{-10}$                    | 16.0  |
| $16 \times 16$ | 2048 | 1.0             | $10^{-2}$  | 0.972            | 0.367               | 857.7               | $9.9981 \times 10^{-3}$       | $-1.85 \times 10^{-6}$                     | 6.9   |
| $16 \times 16$ | 2048 | 1.0             | $10^{-3}$  | 0.972            | 0.370               | $1.249 \times 10^4$ | $9.9999 \times 10^{-4}$       | $-8.67 \times 10^{-9}$                     | 9.7   |
| $16 \times 16$ | 2048 | 1.0             | $10^{-4}$  | 0.972            | 0.370               | $6.347 \times 10^4$ | $1.0000 \times 10^{-4}$       | $-3.35 \times 10^{-10}$                    | 11.4  |

TABLE X.  $\kappa_2(S)$  and  $\lambda_{\min}(W)$  versus grid size at  $\sigma_{\max} = 0.5$  (grids in points;  $n$  is the state dimension). The  $n = 8192$  point was gated off by the projected runtime and memory (Sec. E1).

| Grid           | $n$  | $\lambda^+(H_1)$ | $\epsilon = 10^{-2}$ |                     | $\epsilon = 10^{-3}$ |                     |
|----------------|------|------------------|----------------------|---------------------|----------------------|---------------------|
|                |      |                  | $\kappa_2(S)$        | $\lambda_{\min}(W)$ | $\kappa_2(S)$        | $\lambda_{\min}(W)$ |
| $8 \times 8$   | 512  | 0.6829           | 402.1                | 0.669               | 3009                 | 0.681               |
| $16 \times 8$  | 1024 | 0.6875           | 506.3                | 0.661               | 7009                 | 0.673               |
| $16 \times 16$ | 2048 | 0.6940           | 566.9                | 0.655               | 9856                 | 0.667               |
| $32 \times 16$ | 4096 | 0.6940           | 564.5                | 0.655               | 11241                | 0.668               |

25 candidate approximate symmetrizers  $S_b \approx S$  drawn from six families (Manhattan-distance-banded, magnitude-thresholded, block-pattern, identity-plus-low-rank, low-rank-plus-band hybrid, and banded- $S^{-1}$ ). Each candidate is scored by its residual indefiniteness  $\lambda_b = \lambda_{\max}(\text{Herm}(S_b A_\epsilon S_b^{-1}))$ ; for the exact  $S$ ,  $\lambda_b = -1/(2\lambda_{\max}(W)) < 0$  by Proposition 1. A candidate *passes* if  $\lambda_b \leq 2 \times 10^{-3}$  (the  $\epsilon$ -mode recovery at the default slice remains valid); a candidate with  $2 \times 10^{-3} < \lambda_b \leq 0.5$  is *rescue-eligible*: the classical shift is enlarged from  $\epsilon$  to  $s = \epsilon + \lambda_b$ , so the evolved generator  $S_b(A - sI)S_b^{-1}$  is again dissipative, the undo factor becomes  $e^{sT}$ , and the post-selection penalty relative to the exact  $S$  has exponent

$2\lambda_b T$ . End-to-end recoveries (errors relative to  $e^{AT} z_0$  at  $T \in \{10, 40\}$ ,  $n_p = 10$ ) were run for the exact  $S$  and one representative per surviving family. Table XI lists all candidates on the  $n = 1024$  confirmation grid.

*b. Structure of  $S$ .* Three exact or near-exact structural facts. (i) *Dead-index identity*. On the 328 of 1024 indices (32%; 176 of 512, i.e. 34%, at  $n = 512$ ) that carry no dynamics—interior memory components and the zero padding block— $A$  acts as zero, so Eq. (E1) reduces to  $2\epsilon w = 1$  there and  $S$  is *exactly*  $(2\epsilon)^{-1/2}I$  on this subspace: measured deviation  $2.0 \times 10^{-14}$  on the diagonal, off-diagonal entries below  $7.8 \times 10^{-15}$ , and coupling to the active subspace below  $1.1 \times 10^{-12}$ . This factor is free structure for any implementation. (ii) *The active support is about half the matrix*: the exact  $S$  has 48.1% ( $n = 1024$ ) and 51.1% ( $n = 512$ ) nonzero entries (at threshold  $10^{-14}$  of the largest). (iii) *The correction is full rank but strip-localised*: with the optimal scalar  $\alpha = 16.7$ , the correction  $E = S - \alpha I$  carries 94.5% of the Frobenius norm of  $S$ , has 999 of 1024 singular values above  $10^{-2}$  of the largest, and needs rank 995 to capture 99.99% of its energy—numerically full rank—while 99.6% of that energy sits on index pairs both lying in the absorbing-layer strips (72 of the 1024 indices are strip-free interior).

*c. Findings.* The full-rank measurement rules out identity-plus-low-rank outright: for ranks 8–64 the residual  $\lambda_b = 0.674$ – $0.677$  equals the *untransformed*  $\lambda^+ =$

TABLE XI. All 25 candidate approximate symmetrizers on the  $n = 1024$  ( $16 \times 8$ ) confirmation grid at  $\epsilon = 10^{-2}$ ,  $\sigma_{\max} = 0.5$ . Density is the fraction of nonzero entries of the sparse object; for the  $\alpha I + \text{rank-}r$  family it is  $(r+1)/n$  (the number of LCU terms relative to  $n$ ), and for the banded- $S^{-1}$  family it is the density of the banded  $S^{-1}$  (state preparation by sparse linear solve).  $\lambda_b = \lambda_{\max}(\text{Herm}(S_b A_\epsilon S_b^{-1}))$  is the residual indefiniteness. Status: “pass” =  $\lambda_b \leq 2 \times 10^{-3}$ , valid in  $\epsilon$  mode at the default slice; “rescue” = run end to end with the enlarged shift  $\epsilon + \lambda_b$ ; “elig.” = rescue-eligible ( $\lambda_b \leq 0.5$ ) but not run end to end; “fail” =  $S_b$  indefinite or  $\lambda_b > 0.5$ . The top-50% threshold and the support-of- $A^2$  patterns reproduce the exact  $S$  to machine precision (relative Frobenius deviation  $\leq 6 \times 10^{-16}$ ); the end-to-end errors of every usable candidate are obtained at conditioning  $\text{cond}(S_b) = 5.06 \times 10^2$ – $5.11 \times 10^2$ , equal to the exact value  $5.06 \times 10^2$ .

| Candidate $S_b$                               | density (%) | $\lambda_b$           | status | err. ( $T=10$ )      | err. ( $T=40$ )      |
|-----------------------------------------------|-------------|-----------------------|--------|----------------------|----------------------|
| exact $S$ (active support)                    | 48.1        | $-3.0 \times 10^{-6}$ | pass   | $7.7 \times 10^{-6}$ | $4.9 \times 10^{-5}$ |
| Manhattan band, $d = 1$                       | 1.8         | 2.36                  | fail   | –                    | –                    |
| Manhattan band, $d = 2$                       | 4.2         | 1.32                  | fail   | –                    | –                    |
| Manhattan band, $d = 4$                       | 10.8        | 3.09                  | fail   | –                    | –                    |
| Manhattan band, $d = 8$                       | 26.5        | 0.062                 | elig.  | –                    | –                    |
| Manhattan band, $d = 12$                      | 38.6        | 0.056                 | rescue | $8.7 \times 10^{-6}$ | $3.1 \times 10^{-4}$ |
| magnitude threshold, top 1%                   | 1.0         | 104                   | fail   | –                    | –                    |
| magnitude threshold, top 5%                   | 5.0         | 3.03                  | fail   | –                    | –                    |
| magnitude threshold, top 10%                  | 10.0        | 0.734                 | fail   | –                    | –                    |
| magnitude threshold, top 25%                  | 25.0        | 0.054                 | elig.  | –                    | –                    |
| magnitude threshold, top 35%                  | 35.0        | $4.2 \times 10^{-3}$  | rescue | $8.7 \times 10^{-6}$ | $5.4 \times 10^{-5}$ |
| magnitude threshold, top 50%                  | 48.1        | $-3.0 \times 10^{-6}$ | pass   | –                    | –                    |
| block diagonal (per-field)                    | 7.4         | 6.83                  | fail   | –                    | –                    |
| support of $A$                                | 29.9        | 8.58                  | fail   | –                    | –                    |
| support of $A^2$                              | 48.1        | $-3.0 \times 10^{-6}$ | pass   | $7.7 \times 10^{-6}$ | $4.9 \times 10^{-5}$ |
| $\alpha I + \text{rank } 8$                   | 0.9         | 0.677                 | fail   | –                    | –                    |
| $\alpha I + \text{rank } 16$                  | 1.7         | 0.677                 | fail   | –                    | –                    |
| $\alpha I + \text{rank } 32$                  | 3.2         | 0.674                 | fail   | –                    | –                    |
| $\alpha I + \text{rank } 64$                  | 6.3         | 0.674                 | fail   | –                    | –                    |
| $\alpha I + \text{rank } 32 + \text{band } 2$ | 4.2         | 0.374                 | elig.  | –                    | –                    |
| $\alpha I + \text{rank } 32 + \text{band } 8$ | 26.5        | 0.066                 | rescue | $1.3 \times 10^{-5}$ | $6.6 \times 10^{-4}$ |
| $\alpha I + \text{rank } 64 + \text{band } 2$ | 4.2         | 0.349                 | elig.  | –                    | –                    |
| $\alpha I + \text{rank } 64 + \text{band } 8$ | 26.5        | 0.033                 | elig.  | –                    | –                    |
| banded $S^{-1}$ , $d = 2$                     | 4.0         | 5.02                  | fail   | –                    | –                    |
| banded $S^{-1}$ , $d = 8$                     | 25.3        | 2.23                  | fail   | –                    | –                    |

0.687 to within two percent, i.e. a short LCU correction has no symmetrizing effect at all. Narrow bands, aggressive thresholds, block patterns, and banded inverses are either indefinite or carry  $\lambda_b \gtrsim 1$ . Moderate compression, however, works under the rescue accounting: magnitude thresholding at 35% density recovers to  $8.7 \times 10^{-6}$  ( $T = 10$ ) and  $5.4 \times 10^{-5}$  ( $T = 40$ ); the Manhattan band  $d = 12$  at 38.6% reaches  $8.7 \times 10^{-6}$  and  $3.1 \times 10^{-4}$ ; the rank-32 plus band-8 hybrid at 26.5% reaches  $1.3 \times 10^{-5}$  and  $6.6 \times 10^{-4}$ . The rescue cost is mild at the measured residuals: the largest rescued  $\lambda_b = 0.066$  gives an undo factor  $e^{\lambda_b T} = e^{2.6} \approx 14$  and a post-selection penalty  $e^{2\lambda_b T} = e^{5.3} \approx 2 \times 10^2$  at  $T = 40$ , against the  $T$ -independent baseline  $\kappa_2^2(S) \approx 2.6 \times 10^5$ . The rescue frame is also the robust one: the dissipativity margin of the exact  $S$  is only  $|\lambda_b| = 3.0 \times 10^{-6}$ , and a perturbation of  $S$  of relative spectral norm  $6.4 \times 10^{-7}$  suffices to push  $\lambda_b$  past the  $2 \times 10^{-3}$  gate, so *strict* dissipativ-

ity of any approximation is fragile, whereas absorbing the measured  $\lambda_b$  into the classical shift is exact book-keeping. Taken together—half-dense active support, full-rank strip-localised correction, free dead-index identity factor—these measurements support the implementation verdict of Sec. III D: block-encoding of a  $\sim 35\%$ -dense operator rather than a local circuit or a short LCU, with a structured-ansatz Lyapunov solve as the open problem. The compression study is limited to  $n \leq 1024$ .

## 6. Choosing the shift $\epsilon$

The shift enters the cost twice and the accuracy nowhere: the recovered state is exact up to the classically applied factor  $e^{\epsilon T}$ , while  $\kappa_2(S)$  grows as  $\epsilon^{-0.64} - \epsilon^{-1.03}$  (Sec. E 3) and the evolved norm of the shifted system costs  $e^{2\epsilon T}$  in post-selection. At the recommended

$\epsilon = 10^{-2}$  these are  $\kappa_2(S) = 4.0 \times 10^2 - 8.6 \times 10^2$  across all configurations and  $e^{\epsilon T} \leq e^{0.4}$  ( $e^{2\epsilon T} \leq e^{0.8} \approx 2.2$ ) over the horizons tested ( $T \leq 40$ )—the shift is negligible against  $\kappa_2^2$ , and decreasing  $\epsilon$  by a decade buys only that trivial factor while inflating  $\kappa_2^2$  by one and a half to two decades. The shifted transform is what makes the large- $\epsilon$  choice safe: with the *unshifted* transform of  $A$ , Proposition 1 gives  $\lambda_{\max}(\tilde{H}_1) = \epsilon - 1/(2\lambda_{\max}(W)) \approx \epsilon > 0$ , so the

recovery threshold [Eq. (6)] would demand  $p^* \gtrsim \epsilon T$ —at the default slice  $p^* = 0.117$  ( $n_p = 10$ ,  $p_{\max} = 20$ ) and  $T = 40$  this would force  $\epsilon \lesssim 2.9 \times 10^{-3}$ , excluding the best-conditioned setting, and would couple  $\epsilon$  to the horizon and the  $p$ -register extent. The shifted transform has  $\tilde{H}_1^{(\epsilon)} \preceq -1/(2\lambda_{\max}(W)) I \prec 0$ , so any  $p^* > 0$  is valid at every  $T$  and  $\epsilon$  acts purely as a conditioning knob.

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